

April 2023

Paper I

1. Enumerate bone forming tumors. Imaging features of Osteosarcoma (2+8)
2. Techniques of orbital usg. Appearance of structures in orbital usg. Imaging features of idiopathic orbital inflammation (2+3+5)
3. a) BIRADS classification (5)
b) Role of usg in breast lump (5)
4. Imaging in acute stroke. Role of interventional radiology (6+4).
5. Role of USG in thyroid nodule (10)
6. Neuroimaging of COVID 19 (10)
7. Imaging features of rickets and osteomalacia (5+5)
8. a) Various stages of ICH in MRI. Reason for their appearance. (2+3)
b) Imaging features of congenital TORCH infection (5)
9. Define branchial cleft cyst. Types of branchial cleft cyst. Imaging features. (2+3+5)
10. Types of spinal neoplasms. Imaging in intramedullary spinal lesions (10)

Paper II

1. Bolus tracking technique of MDCT pulmonary angiography. MDCT features of acute pulmonary thromboembolism. (4+6)
2. Anatomy of secondary pulmonary lobule. HRCT features of ILD (3+7)
3. a) Imaging features in Rheumatic heart disease(5)
b) Doppler in evaluation of post operative AV fistula for dialysis (5)
4. a) Color doppler in testicular torsion (5)
b) Lung USG in emergency (5)
5. a) Uterine artery doppler in pregnancy (5)
b) Doppler findings in acute limb ischemia (5)
6. Radiographic features of congenital heart disease. (10)
7. Thoracic manifestations of collagen vascular disease. (10).
8. Types of urinary bladder trauma. Role of imaging and management (4+6)
9. Classification of congenital pulmonary malformations. Imaging features in 3 common. (4+6).
10. Enumerate causes of unilateral opaque hemithorax. Role of imaging in evaluation of unilateral opaque hemithorax (4+6).

Paper III

1. Triple phase hepatic CT protocol. CT features of HCC (5+5)
2. What is intestinal malrotation? Enumerate the types. Radiological features of mid gut volvulus. (2+3+5)
3. Indications of 1st trimester USG. Assessment of fetal anomalies in first trimester USG (4+6)
4. Imaging evaluation and differential diagnosis of post menopausal bleeding in 60 year old female. (4+6)
5. What is fetal growth restriction? Role of USG and doppler in diagnosis and management (2+8)

6. a) Cross sectional anatomy of prostate. MRI in prostate pathologies (2+5)
b) PIRADS (3)
7. Enumerate imaging modalities for adrenal lesions. CT protocol and systematic approach of adrenal lesions (2+3+5)
8. Enumerate biliary tract interventions. Indications, contraindications, procedure and complications of PTBD. (2+8)
9. Imaging in 40 year old female with hematuria. (10)
10. Enumerate causes of lower GI bleed. Role of imaging and interventional radiology in lower GI bleed (3+4+3)

Paper IV

1. Technique of MR neurography and black blood imaging of brain (5+5)
2. Use of isotopes in thyroid lesions (10)
3. Advantages and limitations of 3T over 1.5T MRI. MR safety (6+4)
4. a) Sonoelastography in obstetrics and gynecology (5)
b) Pulmonary MRI (5)
5. a) Hand held radiography (5)
b) Mobile CT (5)
6. Limitations of standard coronary CT with retrospective ECG gating. Mention about current and novel techniques to overcome these limitations
7. Methods of radiation dose reduction in patients in radiography and CT(5+5)
8. Post mortem imaging and applications (10)
9. a) Hepatobiliary MRI contrast agents (5)
b) CT enterography (5)
10. a) Beam restricting devices (5)
b) Thermionic emission in xray production. (5)

December 2022

Paper I

1. Enumerate the various types of spinal dysraphism. Describe the neuroimaging findings in open spinal dysraphism[3+7]
2. Discuss the imaging features of different types of brain herniations and their complications. [7+3]
3. Enumerate the suprahyoid neck spaces. Discuss the contents of the masticator space and diagnostic approach to a mass epicentered in masticator space. [3+7]
4. Describe the radiographic and MR imaging findings in a patient with recurrent shoulder dislocation. Describe the role of MR arthrography in the same. [6+4]
5. Describe the MR protocol for evaluation of temporal lobe epilepsy. Discuss the role of advanced MR imaging and nuclear imaging techniques in evaluation of epilepsy. [5+5]
6. Describe the differentials of an extraconal orbital mass in an 8 year old child along with their imaging findings. [3+7]
7. What are the risk factors for developmental dysplasia of hip? Describe the radiographic and Ultrasound evaluation of developmental dysplasia of hip in a child. [2+4+4]
8. Discuss the differentials along with their imaging findings in a child with retrocerebellar cyst. [10]
9. Discuss the ultrasound and MRI findings in a preterm neonate with hypoxia. [5+5]
10. What are the various causes of Cushing's syndrome? Describe the imaging workup of a patient presenting with Cushing's syndrome. [3+7]

Paper II

1. Discuss the technique and limitations of Focussed Assessment with Sonography in Trauma (FAST). Discuss the extended use of FAST in chest in a case of blunt trauma. [4+(2+4)]
2. Discuss the indications, technique and limitations of cardiac MRI for ischemic heart disease[2+6+2]
3. Enumerate the various causes of unilateral hyperlucent hemithorax on a chest radiograph. Discuss the imaging approach to a child with unilateral lucent lung on chest radiograph. [4+6]
4. Describe the International Thymic Malignancy Interest Group classification of mediastinal compartments. Discuss the imaging differentials of a middle mediastinal mass. [4+6]
5. Discuss the imaging workup of a patient of blunt abdominal trauma presenting with hematuria.[10]
6. Describe the technique for renal doppler. Discuss the role of doppler ultrasound in the evaluation of suspected renal hypertension. [3+7]
7. Describe the colour doppler findings in lower limb venous incompetency. Discuss the interventional procedures for its management. [5+5]
8. Describe the pathophysiology of ground glass opacities in the lung on CT scan. Discuss the differential diagnosis [3+7]

9. Discuss the pathophysiology and associations of tetralogy of Fallot. Describe the radiographic and MRI imaging findings. [(2+2)+(3+3)]
10. Describe the classification of vasculitides. Discuss the imaging features of pulmonary vasculitis [5+5]

Paper III

1. Discuss the complications of acute severe pancreatitis and role of interventional radiology in their management. [6+4]
2. Describe the Liver imaging Reporting and Data system used for imaging findings in a liver lesion, enumerating the major and ancillary features. State the clinical settings in which it is used and others where it is not meant to be used. [7+3]
3. Describe the classic Bosniak criteria for classification of renal cysts. Highlight the clinical significance and drawbacks of this system. [7+3]
4. Enumerate the pretesticular, testicular and post testicular conditions that cause male infertility. Discuss the role of imaging in workup of male infertility. [4+6]
5. Discuss the role of imaging in the workup of a neonate with suspected biliary atresia. What are the imaging differential diagnosis for this condition? [7+3]
6. Describe the radiological anatomy of retroperitoneal spaces. Discuss the imaging approach to a primary retroperitoneal mass lesion. [3+7]
7. Discuss the etiology, imaging findings and interventional management options in hepatic venous outflow tract obstruction. [2+5+3]
8. Discuss the imaging differentials in a 2-year-old child presenting with a liver mass. What are the biochemical investigations that can aid imaging in diagnosis? [7+3]
9. Describe the etiopathogenesis of Hirschsprung's disease. Discuss the imaging features and differential diagnosis. [2+(6+2)]
10. Describe the classification for mullerian duct anomalies. Discuss the imaging features of the different anomalies. [3+7]

Paper IV

1. Discuss the principle and clinical utility of susceptibility weighted MR imaging in brain. [4+6]
2. Describe the various methods to acquire dual energy CT. Discuss the applications of dual energy CT in pulmonary pathologies. [5+5]
3. Compare different types of cerebral MR angiographic techniques discussing their principle, applications and limitations. [4+4+2]
4. Explain the terms Volume CT dose index (CTDI vol) and Dose length product (DLP). Discuss the various techniques for controlling radiation dose in CT. [3+7]
5. Discuss the preoperative imaging evaluation of donor and recipient for liver transplantation. [5+5]
6. Discuss the various types of embolising agents with their clinical use. [5+5]
7. Define impact factor of a journal. Discuss the different types of study designs for research. [2+8]
8. Describe the various radionuclide studies used in evaluation of urinary tract pathology, emphasizing their clinical utility as well as limitations. [10]
9. Describe the construction and principle of an X-ray tube. Discuss causes of X-ray tube failure. [4+3+3]
10. Describe any five ultrasound artifacts along with ways to reduce them and their clinical utility, if any. [2+2+2+2+2]

June 2022

Paper 1

1. 11yr old boy with limping since 1 mn - a. Radiological role
b. Causes and imaging
2. Leukocoria causes and imaging.
3. Salivary gland tumors and imaging.
4. Malignant thyroid lesion and imaging.
5. Causes of SAH and imaging
6. Cystic lesions of jaw and imaging.
7. CNS manifestation in HIV with imaging.
8. Skeletal findings in leukemia and hemolytic anemia
9. Meniscal tear MRI and MR arthrogram role.
10. Nerve sheath tumors and imaging

Paper II

1. triple cut CT coronary angiography technique. its principle and protocol.
2. role of radiology in erectile dysfunction.
3. vascular anomaly international classification and imaging findings.
4. color doppler artefacts and how to prevent it.
5. multinodular pulmonary lesions and its imaging.
6. cystic lesions of lung and imaging
7. USG in pleural and parenchymal pathology.
8. Right pelvic pain in a woman causes and imaging.
9. hemoptysis role of radiology and intervention.
10. cardiac tumors and imaging.

Paper III

1. role of imaging in management of renal transplant patient.
2. role of imaging in chronic liver disease.
3. 74 yr old with urinary incontinence - approach and role of imaging.
4. cystic pancreatic tumors and imaging.
5. usg in ectopic pregnancy.
6. causes of nephrocalcinosis and imaging.
7. pathophysiology, xray , usg and CT in genitourinary TB.
8. hyperenhancing tumors of liver and imaging.
9. imaging of necrotizing enterocolitis.
10. imaging in mesenteric ischemia

Paper IV

1. what is k space. various techniques using it.
2. elastography - principle , technique and applications
3. scatter radiation and ways to reduce it.
4. chemical shift imaging in mri. technique , advantages and disadvantages.
5. a) pulse fluoroscopy b) zero lead apron.
6. various fat suppressed sequence. describe them.
7. MR perfusion techniques.
8. parallel imaging and its use.
9. a. breast tomosynthesis b. artificial intelligence
10. MR urography. technique. application, adv and disadvantages.

Dec 2021

Paper I

1. Manifestations of CNS TB. Clinical findings, patho, imaging of any 2
2. Midline pediatric neck masses. Embryology, clinical & imaging thyroglossal cysts
3. Diffuse hyperostosis skull. Imaging of any 2
4. 60 Y female breast masses d/d. Benign vs malignant on USG & MRI
5. A. Perthes disease.
B. Ewing sarcoma femur.
6. Exophthalmos causes. Role of imaging.
7. Internal derangement of knee causes. MRI knee protocol. ACL tear findings.
8. A. Sella turcica anatomy. Masses within and adjacent to it.
B. Imaging in NF2.
9. A. CT in ischemic stroke
B. Sturge Weber syndrome
10. A. Cleidocranial dysplasia
B. Rickets

Paper II

1. Radiographic compartments of mediastinum.
Pediatric mediastinal masses.
Posterior mediastinal masses in 1 yr old.
2. Pathophysiology of COVID pulmonary infection. Chest CT protocol. Chest CT findings.
3. Causes of hilar node enlargement. How to differentiate causes on imaging.
4. Lifecycle of Echinococcus granulosus.
Pulmonary findings in hydatid cyst.
5. Imaging protocol in 50 yr old with acute chest pain.
Likely causes and imaging findings.
6. Any 5 chest radiographic views. Technique and indications.
7. A. Imaging in varicose veins
B. MR in ischemic heart disease.
8. A. Radiographic findings in congenital cyanotic heart disease.
B. Imaging in chronic PTE
9. A. Differentials of calcifications on chest X-ray.
B. Differentials of cavitary pulmonary lesions
10. A. Imaging findings mitral stenosis
B. Imaging in renal artery stenosis

Paper III

1. Various benign and malignant solid renal masses. Imaging features of malignant solid renal masses.
2. Draw and discuss
A. T staging of bladder tumors
B. Male urethral injury
3. Role of USG, CT and MRI in endometrial CA.

4. Role of virtual colonoscopy. Technique and contraindications. Imaging findings of colonic polyps on virtual colonoscopy.
5. Role on USG and MRI in morbidly adherent placenta.
6. A. Bleeding PV at 18 weeks. Causes and imaging findings.
B. Oligohydramnios
7. A. Causes of dysphagia. Role of barium swallow.
B. MR enteroclysis.
8. A. MRI in ca cervix.
B. Antenatal MRI - indications, safety and protocols
9. A. MRCP
B. Caroli's disease
10. A. X-ray findings in acute pancreatitis.
B. Role of CT in initial imaging and follow up of GIST

Paper IV

1. Labelled diagram of x-ray tube. Functioning of x-ray tube. Causes of tube failure.
2. A. X-ray film.
B. Computed radiography.
3. Imaging techniques for cartilage imaging. Common sites for osteochondral defects and their imaging appearances.
4. A. Ultrasound contrast agents.
B. CT enteric contrast agents.
5. Techniques and role of interventional management in malignant liver tumors.
6. Radiation dose limits in radiation workers and general people. Radiation protection measures for A. Staff. B. Patient. C. General population.
7. A. ALARA and eLORA
B. HIFU - Principle and clinical uses
8. A. PET CT in oncology.
B. Contrast reaction management in CT unit.
9. A. Principle of color doppler imaging.
B. Principle and pitfalls of TOF MRA.
10. A. Scintigraphy in cardiac imaging.
B. Diffusion weighted imaging

Previous years question papers

June 2021

Paper I

- 1) Role of various imaging modalities in evaluating HIE.
Imaging findings in HIE.
Preterm HIE vs full term HIE.
- 2) Renal osteodystrophy - imaging findings.
Approach to imaging in hyperparathyroidism.
- 3) Latest nomenclature of disc herniation.
Osteoporotic vertebral collapse vs Metastatic vertebral collapse.
- 4) Indications of MRI breast. Technique of MRI breast. Non-enhancing mass in MRI breast.
- 5) Ultrasound and MRI findings of rheumatoid arthritis.
- 6) Radiological anatomy of central skull base. and Imaging in CSF rhinorrhea.
- 7) Imaging features of recurrent shoulder dislocation.
Role MR arthrography in shoulder displacement.
- 8) Imaging features and demographics of craniopharyngioma
Discuss the it's differential diagnosis.
- 9) Suprahyoid neck spaces - boundaries and Radiological anatomy.
Pathologies of masticator space.
- 10) Radiological anatomy of ligaments of knee joint.
MR evaluation of cartilages of knee joint.

Paper II

- 1) CT and radiographic anatomy of bronchopulmonary segments. Hilum overlay sign and hilum convergence sign.
- 2) Pathologies of lung carcinoma.
TNM8 staging in lung carcinoma.
FDG PET role in lung carcinoma.
- 3) ITMIG classification of mediastinum.
Thymoma- imaging features and approach.

Imaging features thymoma vs hyperplasia.

4) Pelvic fractures - classification and Imaging approach.

Osseous and soft tissue injury patterns of pelvic fractures.

5) Solitary pulmonary nodule definition.

Discuss the etiologies/differential diagnosis.

Benign vs malignant SPN.

6) Dark and white blood cardiac MRI.

Viability scans in IHD.

7) MDCT reconstruction methods in the airway.

Imaging intraluminal lesions of tracheo-bronchial tree.

8) Constrictive pericarditis - CT imaging.

Radiographic Imaging of mitral valve diseases.

9) Abdominal trauma- imaging approach.

Splenic injury Imaging.

10) Peripheral arterial disease - imaging with advantages and disadvantages.

Paper III

1. (a) CT imaging in pancreatitis.

(b) what is groove pancreatitis?

(c) Radiological difference between focal mass forming pancreatitis and pancreatic adenocarcinoma

2. (a) LIRAD, its main features and categories

(b) use of LIRAD in liver lesion management

3. (a) imaging features of intestinal TB on barium meal follow through and CT (b) difference between intestinal TB and crohns disease

4. (a) Radiological anatomy of retroperitoneum

(b) cross sectional imaging in retroperitoneal liposarcoma

5. (a) Bosniak classification of renal cyst 2019

(b) syndromic association of renal cyst in adult

6. different variety of gestational trophoblastic disease and role of imaging

7. (a) imaging evaluation of female presented with infertility

(b) mri imaging feature of uterine adenomyosis

8. (a) MRI evaluation method in ovarian mass

(b) differentiating features of benign and malignant ovarian mass on USG and MRI

9. (a) Imaging in small bowel lymphoma
(b) Discuss differentials
10. MRI hepatobiliary contrast agent and its use in differentiating liver lesion

Paper IV

1. (a) brief principle of DWI MRI
(b) 4 brain pathology showing restriction
(c) non-neurological application
2. motion artifacts in MRI. Methods to prevent them.
3. (a) What is Nephrogenic systemic fibrosis
(b) Risk factors for NSF
(c) method of prevention
4. principle of ultrasound elastography and its application in diffuse liver disease
5. (a) principle and types of radiographic grid.
(b) what is grid ratio
(c) advantages and disadvantages
6. (a) What are the stochastic and nonstochastic effect of radiation ? (b) AERB dose limits for general public and occupational worker
7. First and second generation of ultrasound contrast agent and its application
8. (a) cone beam CT
(b) Zoning in MRI
9. (a) Principle of RFA
(b) application in benign and malignant lesions
10. technique, indications, contraindications and complications of HSG

December 2020

Paper IV

1. Principle of dual energy CT and its various applications.
2. Technique, imaging sequences and clinical application of MR urography.
3. Describe briefly various embolizing agents used in interventional procedures.
4. a) Rotating anode tube.
b) Photoelectric effect.

5. Contrast reactions and its management.
6. a) Rare earth screens.
b) TLD badge
7. a) Anatomy of bronchial artery.
b) Indications, technique and complications of bronchial artery embolization.
8. Describe image intensifier with the help of a diagram and briefly discuss its properties and limitations.
9. a) Tissue harmonic imaging.
b) MR artefacts.
10. Recent developments in non-invasive lymphatic imaging techniques for visualizing the central lymphatic anatomy and flow dynamics.

Paper III

1. a) Enumerate the causes of acute pelvic pain in a young female patient.
b) Role of imaging in ovarian torsion.
2. a) Describe various causes and role of imaging modalities in male infertility.
b) Role of sonography in erectile dysfunction
3. Enumerate various causes of post-menopausal bleeding in a 60-year female. Discuss the role of imaging in evaluation of endometrial carcinoma and its management based on staging.
4. a) Radiographic, ultrasound and MDCT findings in acute mesenteric ischemia.
b) Role of intervention in this condition.
5. Draw schematic diagram of anorectal junction on the coronal plane, depicting internal and external sphincters. Describe the types of anorectal fistula with the help of a diagram and briefly discuss MR imaging protocol in their evaluation.
6. Role of ultrasound, MRI, and scintigraphy in assessment of biliary atresia.
7. Name various motility disorders of esophagus. Discuss pathophysiology and imaging findings of cardia achalasia.
8. Discuss Imaging features of renal artery stenosis on ultrasonography and MR angiography and briefly the role of interventions in its management

9. Define pancreatic divisum. Briefly discuss embryological basis with the help of diagrams. Describe imaging findings on MDCT, MRCP, ERCP.
10. Describe approach to a patient in renal trauma with a flow chart & discuss the role of imaging with grades of renal injury.

Paper II

1. a) Enumerate causes of inferior rib notching.
b) Describe X-ray, CT and MRI findings in Coarctation of aorta.
c) Briefly discuss role of interventional radiology in its management.
2. a) Radiological patterns of pneumonias on chest computed tomography in immunocompromised patients.
b) Imaging features of fungal infections.
3. Enumerate the causes of hemoptysis in adult patient. Briefly discuss indications, techniques and complications of radiological interventions in them.
4. a) Arterial and venous anatomy of lower limb with help of diagram.
b) Imaging features of deep venous thrombosis.
5. Define acute aortic syndrome. Classify aortic dissection. Describe its imaging features on CT angiography and its management.
6. Role of nuclear imaging, cardiac MRI and MDCT in evaluation of ischemic heart disease patients.
7. Enumerate causes of neonatal respiratory distress. Describe in detail pathophysiology and imaging findings in hyaline membrane disease.
8. Draw a schematic diagram describing anatomy of diaphragm and imaging features of diaphragmatic injury on chest X-ray, ultrasound, CT and MRI.
9. Describe the anatomy of mediastinum with the help of a diagram. Discuss imaging features of anterior mediastinal masses.
10. Define pulmonary edema and describe its pathophysiology. Enumerate causes and imaging features in pulmonary edema.

Paper I

1. a) Imaging evaluation of a patient with Cushing's syndrome.
b) Role of imaging in differentiating adrenal adenoma from adrenal metastasis.
2. a) Approach for incidental thyroid nodules detected on imaging.
b) TI-RADS and its clinical utility.
3. What is lipoma arborescens? Enumerate imaging findings on US and MRI. Discuss any two differential diagnoses.
4. Indications, technique, advantages and limitations of MRI in imaging of breast cancer.
5. Discuss the etiology of femoroacetabular impingement and imaging findings on plain radiographs and MRI.
6. Enumerate the various posterior fossa tumours in the paediatric age group. Describe in brief the protocol for evaluation of these patients. Discuss the salient imaging features of the three commonest paediatric posterior fossa tumours.
7. Discuss in brief the salient CT and MRI features in an acute ischemic stroke patient arriving at the hospital within 6 hours of onset. Suggest an algorithm based approach for confirmation of diagnosis and guiding the management of such a patient.
8. Describe the imaging protocol of paranasal sinuses for FESS (Functional Endoscopic Sinus Surgery). What are the important anatomical landmarks and anatomical variants which should be looked for while evaluating for FESS? Discuss the radiological patterns and features of Aspergillus infection of paranasal sinuses.
9. a) Discuss the radiological anatomy of ankle ligaments.
b) Briefly state various MR sequences and planes used to evaluate ankle joint on magnetic resonance imaging.
10. A 37-year-old female presents with a history of intractable epilepsy. Discuss the imaging modalities and protocols used in workup of this patient. Describe imaging findings of mesial temporal sclerosis.

June 2020

Paper I

1. Discuss in brief, the salient CT and MRI features in an acute ischemic stroke patient arriving to the hospital within 6 hours of onset. Suggest an algorithm based approach for confirmation of diagnosis and guiding the management of such a patient.
2. A young adult patient has presented with unilateral pulsatile proptosis and chemosis after a road traffic accident. What is the probable etiology and how will you confirm the diagnosis by imaging? Discuss the role of radiologic Interventions in managing such a condition.
3. Discuss in brief, various types of hyperparathyroidism. Describe imaging features of various types of hyperparathyroidism.
4. Discuss various indications, techniques, advantages and limitations of MRI in imaging of breast cancer.
5. Describe various morphological patterns of vertebral body injuries in thoraco-lumbar spine. Discuss role of MRI in the diagnosis and management of such vertebral body injuries.
6. Enumerate various causes for unilateral painful hip in a 10-year old child. Discuss the role of imaging in any two pathologies.
7. a) Describe briefly various neurocutaneous syndromes.
b) Describe imaging features of von Hippel Lindau syndrome.
8. Discuss the role of HRCT and MRI in the evaluation of congenital sensorineural hearing loss.
9. Discuss basic principles of MR Spectroscopy. Discuss in brief, role of MR Spectroscopy in the evaluation of various brain pathologies.
10. A 14-year-old male presented with fever, pain and swelling of right proximal arm.
a) What are the various differential diagnoses?
b) Discuss in brief, role of imaging in the evaluation of such a case.

Paper II

1. Describe in brief, protocol for CT pulmonary angiography for a suspected case of pulmonary thromboembolism. Describe plain radiographic & CT features of pulmonary thromboembolism.

2. Describe the imaging protocol and role of MRI in evaluation of ischemic heart disease.
3. a) Hilum overlay sign and its significance on diagnostic radiology.
b) Cervicothoracic sign and its significance on diagnostic radiology.
4. Describe various types of aortic dissection. Discuss the role of imaging in evaluation of aortic dissection.
5. Describe various types of lung collapse. Discuss radiologic features of collapse in general and the specific imaging features which help in localization of collapse.
6. Discuss the indications of imaging in a suspected COVID-19 patient. Describe imaging features and pattern of involvement of respiratory system in COVID-19.
7. Discuss various imaging signs of diaphragmatic injury. Describe the differences between blunt and penetrating diaphragmatic injury. Discuss signs to differentiate diaphragmatic hernia from eventration and paralysis.
8. Discuss the radiological anatomy of coronary arteries. Describe the technique of CT coronary angiography. Discuss the recent advances in assessment of coronary artery stenosis by CT angiography.
9. Discuss the imaging approach to acute limb ischemia. Discuss the interventional radiological treatment options for it.
10. Discuss the imaging evaluation of various catheters, tubes and lines seen on a chest radiograph in an ICU setting.

Paper III

1. Discuss the causes and radiological features of Budd Chiari Syndrome. Discuss the role of Interventional Radiology in treatment of Budd Chiari Syndrome.
2. a) Imaging workup of a potential renal donor.
b) Imaging evaluation of vascular complications of a transplant kidney.
3. Enumerate various internal hernias. Discuss in detail imaging features of any two internal hernias.
4. Describe imaging algorithm in a suspected ectopic pregnancy case. Describe role of imaging in differentiating three common clinical mimics of ectopic pregnancy.

5. Discuss the classification and imaging features of cystic neoplasms of pancreas.
6. Describe TNM staging of bladder cancer. Draw a schematic diagram of T staging of bladder tumour. Enumerate the role of ultrasound, computed tomography and MRI in the diagnosis and staging of the bladder carcinoma.
7. Describe various indications, protocol, advantages and disadvantages of fetal MRI.
8. What are the various causes of acute mesenteric ischemia? Discuss the role of radiology in its diagnosis and management.
9. Describe briefly segmental anatomy of the liver with a line diagram. Describe imaging (CT & MRI) features in hepatocellular carcinoma.
10. a) Role of plain radiograph in a case of acute abdomen.
b) Role of MRI in acute pancreatitis.

Paper IV

1. a) Rotating anode X-ray tube and its advantages.
b) X-ray beam restricting device and their uses.
2. Classify water soluble iodinated contrast media. Describe various adverse reactions to iodinated contrast media. Describe steps in the diagnosis and management of contrast media induced anaphylaxis.
3. Enumerate the principles of Computed Radiography (CR) and Digital Radiography (DR). What are their advantages and disadvantages?
4. a) Describe the principle and technique of phase contrast MR angiography.
b) Discuss its advantages, limitations and clinical applications.
5. a) Applications of dual energy CT.
b) Ultrasound artifacts.
6. Classify the commonly used embolizing agents. Discuss their advantages, disadvantages and clinical applications.
7. What is plagiarism in scientific writing? What are the various common types of plagiarism? Describe how plagiarism can be detected in scientific writings. Briefly mention a few measures to avoid plagiarism.
8. a) Predatory journals and publishing.

- b) Techniques of randomization of samples in a research study.
- 9. a) PC-PNDT Act and its applications.
- b) TLD Badge: Principles and uses.
- 10. a) Artificial Intelligence in radiology.
- b) PET MRI.

December 2019

Paper I

1. a) Draw a labelled diagram of neck spaces. 6+4
b) Enumerate lesions of parapharyngeal spaces.
2. a) Describe in brief revised WHO classification of brain tumours. 7+3
b) Enumerate primitive Neuroectodermal tumours of the brain.
3. a) A 45-year-old female came with a lump in unilateral breast. How will you evaluate this patient on imaging? 6+4
b) Describe BI-RADS classification used for reporting of mammogram.
4. a) Enumerate various causes of hemolytic anemia. 2+(5+3)
b) Describe the imaging findings in a case of thalassemia major. How will you differentiate it from sickle cell disease on imaging?
5. a) State the distinguishing features of intramedullary, extramedullary-intradural and extradural spinal lesions on MRI. 6+4
b) Discuss briefly the differential diagnosis of intramedullary spinal lesions.
6. a) Describe in brief anatomy of the shoulder joint. 4+6
b) Briefly state various MR sequences and planes used to evaluate shoulder joint on MR (Magnetic resonance) imaging.
7. a) What are the causes of hyperparathyroidism? 4+6
b) Enumerate the finding on Plain film, CT and scintigraphy of primary hyperparathyroidism.
8. a) Enumerate causes of unilateral proptosis. 4+(3+3)
b) Describe briefly imaging findings of optic gliomas and Caroticocavernous fistula.
9. A 40 year old female has presented with loss of vision and instability in gait. 6+4
a) Discuss the differential diagnosis and MR findings of the most probable cause.
b) What is the role of Diffusion Tensor Imaging in this patient?
10. A 15-year boy presents with localized swelling and pain of 2 months duration involving right lower thigh. 4+6
a) What will be its differential diagnosis?
b) Discuss Conventional Radiographic, CT and MRI features of the commonest primary malignant bone tumour in this age.

Paper II

Write Short notes on:

PART A

1. a) Describe anatomy of lung on High Resolution Computed Tomography (HRCT). 3+3+4
b) What are the common patterns of imaging features seen in various interstitial lung diseases (ILD)?
c) Describe HRCT features of Usual Interstitial Pneumonia (UIP).
2. A 25-year-old presented with life threatening hemoptysis. 2+(4+4)
a) Draw an algorithm to outline radiological management of such a case.
b) Discuss in brief imaging features on conventional radiology, CT scan and MRI; and role of interventional radiology.
3. a) Describe the arterial anatomy of carotid vascular system with the help of labelled diagram. 4+6
b) Discuss the role of Ultrasound and Color Doppler imaging in evaluation of extra cranial carotid occlusive disease.
4. a) What do you understand by the term extramedullary hematopoiesis? 2+3+5
b) Enumerate its causes.
c) Discuss its plain film and cross sectional imaging findings.
5. a) Enumerate the causes of acute respiratory distress syndrome. 5+5
b) Give in detail radiological management of aortic dissection.

PART B

6. a) Enumerate causes of thoracic aortic aneurysm in a patient below the age of 40. 3+3+4
b) Briefly describe the role of imaging in the diagnosis of these patients.
c) Describe radiological management of these patients.
7. a) Describe the normal anatomy of coronary arteries. 4+6
b) Discuss the role of MDCT in coronary artery diseases.
8. a) Enumerate various tumours of heart. 5+5
b) Describe the imaging features in myxoma of heart.
9. a) Discuss the differential diagnosis in a 7-year old boy presenting with a pulsatile swelling of the right forearm. 4+6
b) Describe the imaging findings with Doppler and MRI.
10. a) Venous anatomy of lower limb with the help of a diagram. 4+6
b) Technique of Color Doppler Imaging of lower limb veins and imaging features of deep venous thrombosis (DVT).

Paper III

1. a) Describe the anatomy of retroperitoneum on imaging. 4+6
b) Describe the radiological findings of 2 common non- tumour conditions occurring in the retroperitoneum.
2. a) Name the diseases associated with H.pylori infection. 4+6
b) Briefly discuss Barium meal features of benign and malignant gastric ulcers supported by a suitable diagram.
3. a) Mention the various interventional techniques used in the management of hepatocellular carcinoma (HCC). 4+6
b) Briefly describe the guidelines outlining the criteria of response assessment of hepatocellular carcinoma.
4. a) What are the causes of vesico-ureteric reflux? 5+5
b) Describe the imaging in vesico-ureteric reflux along with its grade with well labelled diagrams.
5. a) Enumerate causes of Gastroduodenal artery bleed. 3+3+4
b) Mention current imaging techniques for its diagnosis.
c) Describe Interventional Radiology techniques used in its management.
6. a) Enumerate causes of painless hematuria in a 55-year-old patient. 4+(3+3)
b) Discuss the imaging features and role of Interventional Radiology in any two pathologies.
7. a) Enumerate various complications of acute and chronic pancreatitis. 3+3+4
b) Describe in brief their imaging features.
c) Describe in brief their management by Interventional Radiologist.
8. a) Enumerate various color Doppler parameters used in IUGR. 4+4+2
b) Briefly discuss their role in IUGR.
c) Mention the significance of aortic isthmus index.
9. A 36-year-old male presented to emergency department with blunt abdominal trauma. 5+5
a) Describe in brief imaging findings requiring urgent intervention.
b) Discuss MDCT findings in hepatic injury.
10. a) Describe penile circulation. 3+3+4
b) What are the causes of male impotence?
c) Role of Color Doppler imaging in impotence.

Paper IV

- | | | |
|-----|--|---------|
| 1. | Discuss indications, technique, management and complications of bronchial artery embolization. | 2+2+3+3 |
| 2. | a) Enumerate various ultrasonic contrast media.
b) Describe their principle.
c) Describe in brief their clinical application in evaluation of liver mass lesions. | 4+3+3 |
| 3. | a) Discuss the role of Scintigraphy in cardiac imaging.
b) Describe various artifacts and interpretation pitfalls associated with this technique.
c) Describe its role in brief in myocardial perfusion and viability. | 4+3+3 |
| 4. | a) Describe principles and techniques of USG Elastography.
b) Describe in detail clinical applications of compression Elastography. | 6+4 |
| 5. | a) Describe the basis of BOLD Imaging technique.
b) Describe its utility and limitations. | 4+6 |
| 6. | a) Describe MR contrast media for liver imaging.
b) What is contrast induced nephropathy (CIN)?
c) What will you do to prevent CIN? | 4+3+3 |
| 7. | a) Describe the principle of Dual Energy CT.
b) Describe its various applications. | 5+5 |
| 8. | a) What is p value? What is its significance and clinical application in research?
b) What is sensitivity, specificity, positive predictive value, negative predictive value and accuracy? | 3+7 |
| 9. | a) What do you mean by AERB, eLORA and ALARA?
b) How will you minimize Radiation Hazards? | 4+6 |
| 10. | a) Artificial Intelligence in Health care.
b) Discuss its advantages and disadvantages. | 5+5 |

June 2019

Paper I

1. Describe the anatomy of the larynx with the help of diagrams. How does radiology help to assess staging of laryngeal malignancy?
2. Role of imaging in short stature.
3. What are the CNS infections in HIV? Discuss their neuroimaging characteristics.
4. Enumerate the different extraconal lesions of the orbit. Briefly mention their imaging findings.
5. Radiological assessment of CSF rhinorrhea.
6. What are the various forms of Langerhans Cell Histiocytosis (LCH)? Describe the radiological features of the imaging spectrum.
7. Describe the anatomy of cranio-vertebral junction with the help of diagram(s). What are the various radiological measurement parameters to diagnose Basilar Invagination (BI)?
8. Discuss the various techniques of cartilage imaging. Describe the common sites and imaging features of osteochondral lesions.
9. Which are the common sites for occurrence of Brown tumours? How will you evaluate the patient to detect the underlying abnormality?
10. Describe the BI-RADS lexicon pertaining to MR Mammography. Describe the MR Mammography findings in a BI-RADS 5 breast lesion.

Paper II

1. How do you classify aortic aneurysms? Discuss the role of radiological intervention in abdominal aortic aneurysms. What can be the possible complications post intervention?
2. Pathogenesis and the spectrum of plain radiography findings in rheumatic heart disease.
3. Radiological spectrum of pulmonary Aspergillosis.
4. Role of radiology in diagnosing and treating renovascular hypertension.
5. Radiology of posterior mediastinal masses in children

6. Discuss the various imaging features of trauma to the chest. What are the imaging findings in broncho-pleural fistula?
7. Classify interstitial lung disease and give the relevant HRCT findings in each.
8. What are the indications for cardiac evaluation of “delayed hyper-enhancement” (DHE) in MRI? Which MRI sequences are usually needed for DHE assessment? Describe the cardiac MRI features of Hypertrophic Obstructive Cardiomyopathy (HOCM).
9. Tabulate the grading of liver trauma. What are the important delayed complications of injury to the liver?
10. Describe the common variations in anatomy of aortic arch branches.

Paper III

1. Etiology and characteristic imaging findings of retroperitoneal fibrosis.
2. Mention causes of solitary hepatic space occupying lesions. How can MRI help in differentiating them?
3. A patient present with 10 weeks amenorrhoea followed by bleeding per vaginum. What are the possible causes and how will you evaluate the patient radiologically?
4. Role of imaging in acute pancreatitis.
5. What are the sites of origin of Pheochromocytoma? Describe the imaging features of this tumour and its differentials.
6. What are the causes of an acute abdomen in a neonate? What are imaging features of necrotizing enterocolitis?
7. What are the sites of physiological indentation of the esophagus? Discuss radiological diagnosis of achalasia cardia and its complications. What is the role of radiological intervention?
8. Pathogenesis of genito-urinary tuberculosis and its imaging findings in CT scan.
9. Mention the various causes of infertility in male. Describe the role of imaging in the evaluation of erectile dysfunction.
10. Role of Doppler in pregnancy.

Paper IV

1. Draw a diagram of the hepatic segments. Describe the role of radiology in the work-up of a liver transplant.
2. Enumerate the indications of Percutaneous Nephrostomy (PCN). Describe the technique.
3. Radiofrequency Ablation (RFA) in liver masses - the physics and the technique.
4. Describe the AERB guidelines to set up a Radiology centre for a single static X-ray machine. Supplement it with a diagram.
5. What are various methods available for dose reduction in CT? Discuss the technique of CT perfusion imaging.
6. What is an MRI fringe field? How can the MRI safety risks be minimized in your department?
7. Classify iodinated contrast media used in diagnostic imaging. What steps will you take to prevent contrast induced nephropathy?
8. What are the important parameters of image quality in digital radiography? What are the advantages of digital radiography over conventional radiography?
9. Describe MRI contrast media. What is a Nano diamond?
10. Mention the basic physics behind Positron Emission Tomography (PET). Describe the role of PET in staging of lymphoma.

December 2018

Paper I

1. Enumerate the various causes of bilateral T2 hyper intense lesions in basal ganglia. Describe the neuroimaging findings in Wilson's disease.
2. a) Role of magnetic resonance imaging in head injury.
b) Role of diffusion weighted imaging in intracranial infections.
3. a) Draw a labelled diagram of a coronal section through the "cavernous sinus".
b) Discuss the differential diagnosis of cavernous sinus lesions.

4. Draw a labelled diagram depicting the normal anatomy of orbit. Discuss the imaging findings of orbital pseudotumor and thyroid ophthalmopathy.
 - a) Computed tomographic and magnetic resonance imaging findings in mycotic sinusitis.
 - b) Imaging features of orbital involvement in Neurofibromatosis type 1.
6. Depict the anatomy of maxillary sinus using a labelled diagram. Discuss the role of imaging in the diagnosis of the mucocoele of maxillary sinus.
7. a) Explain the role of HRCT temporal bone and MRI in the work-up of a case of congenital sensorineural hearing loss (SNHL) for a possible cochlear implant.
 - b) Enumerate the key CT imaging features which must be commented upon from the perspective of a surgeon about to undertake a cochlear implant.
8. Explain what you understand by the term BIRADS. Enumerate the indications of magnetic resonance mammography. Explain the technique of magnetic resonance mammography.
9. Enumerate the causes of osteoporosis. Discuss the role of plain radiography in the evaluation of skeletal manifestations of hyperparathyroidism.
10. a) Explain the role of plain radiography in various diseases of the joints.
 - b) What is the role of ultrasound of joints in the evaluation of a case of rheumatoid arthritis?

Paper II

1. Enumerate the causes of bilateral lung cysts in an adult and discuss the role of HRCT in the differential diagnosis.
2. Enumerate the causes of lobar collapse. Describe the various chest radiographic findings of left upper lobe and left lower lobe collapse.
3. Draw a line diagram to define the radiological anatomy of mediastinum. Discuss the differential diagnoses of anterior mediastinal masses based on computed tomography examination findings.
4. Enumerate the causes of bilateral upper lobe fibrosis. Discuss the HRCT findings in intrathoracic sarcoidosis.

5. Enumerate the various lines and tubes encountered on chest radiographs in patients admitted to the intensive care unit (ICU). Discuss the imaging findings in pulmonary oedema.
6. a) Describe the technique of multi-detector CT pulmonary angiography (MDCTPA) in a suspected patient of acute pulmonary thromboembolism.
b) Enumerate the false positive and false negative CT Pulmonary Angiography (CTPA) findings which can influence the assessment in such a patient suspected with pulmonary thromboembolism.
7. a) Myocardial viability imaging.
b) Chest radiograph findings in mitral valve disease
8. a) Imaging features in Tetralogy of Fallot.
b) Imaging features in partial anomalous pulmonary venous drainage.
9. Enumerate the benign pulmonary tumours and discuss their differential diagnosis.
10. Outline the following using an algorithmic sequence:
a) Imaging approach in a patient with blunt trauma to chest.
b) Imaging approach in a patient with blunt trauma to abdomen.

Paper III

1. Describe the role of conventional radiography and computed tomography in the evaluation of a suspected case of acute large bowel obstruction.
2. a) Imaging in esophageal motility disorders.
b) Imaging in postoperative stomach.
3. Enumerate the various causes of strictures in the small intestine. Describe the imaging findings in Crohn's disease of the small intestine.
4. a) Role of magnetic resonance imaging in the evaluation of hemochromatosis and iron overload states.
b) Role of sonographic and Doppler evaluation in a patient of portal hypertension.
5. Describe the technique of multiphasic CT scan study of the liver. Discuss the imaging findings of hepatoma
6. Describe the indications, contraindications and technique of CT urography.

7. Describe the role of ultrasound, computed tomography and magnetic resonance imaging in the diagnosis and staging of endometrial carcinoma.
8. Describe the indications, contraindications, technique and complications of hysterosalpingography.
9. a) PCPNDT Act.
b) Ultrasound findings in a 12 week foetus with Down syndrome.
10. a) Imaging findings in abruptio placentae.
b) Imaging findings in molar pregnancy.

Paper IV

1. Discuss the physical principles of the ultrasound contrast media. Enumerate the safety issues and recommendations regarding their clinical use.
2. Define 4DCT technique and enumerate the situations in which it is used and discuss in detail the technique of 4DCT of parathyroid.
3. Discuss the factors which tend to affect the image quality on a computed tomography (CT) scan. Describe the diverse radiation dose reduction strategies which can be employed while carrying out a thoracic CT scan.
4. Enumerate the various post-processing techniques which can be used during a thoracic computed tomography (CT) examination. State the clinical utility of each of these postprocessing techniques citing examples.
5. Enumerate the various pulse sequences used in magnetic resonance imaging. State the clinical utility of each pulse sequence citing examples.
6. Describe the basic principles of PET-CT. Discuss the advantages and disadvantages of PET-MRI over PET CT.
7. Enumerate the various liquid embolic agents used in interventional Radiology. Discuss the role of interventional Radiologist in the management of Cerebral Arteriovenous malformations.
8. Discuss the indications, pharmacokinetics, and method of iodine-131 metaiodobenzylguanidine scintigraphy.
9. Describe the basic physical principles, indications, contraindications and technique of radiofrequency ablation of liver tumours.
10. a) Kappa measure (in terms of statistical analysis of medical data)
b) The Receiver Operating Characteristics (ROC).

June 2018

Paper I

1. a) Magnetic resonance imaging findings in a patient suspected to have acute ischemic cerebral stroke.
b) Endovascular treatment in acute ischemic cerebral stroke as per the current evidence.
2. Enumerate the various types of primary traumatic lesions in head injury. State the indications for computed tomography (CT) and that for magnetic resonance imaging (MRI) in head trauma.
3. A middle aged lady presents with a breast mass. What are the possibilities? Discuss the specific mammographic characteristics which can help distinguish between the benign and malignant breast lesions.
4. Enumerate the cystic lesions of the jaw. Outline the radiological algorithmic approach towards the diagnosis of such a lesion.
5. Enumerate the various labral injuries of the shoulder. Describe the magnetic resonance imaging findings in any two such injuries.
6. Enumerate the possible causes of a jugular foramen mass. Describe the magnetic resonance imaging features in any two such lesions.
7. Enumerate the causes of acute scrotal pain. Discuss the specific sonographic findings which will come in useful in making the diagnosis.
8. Enumerate the various magnetic resonance imaging (MRI) sequences which can help in the evaluation of a cerebral tumour. Describe the key MRI features of a high grade glioma highlighting the significance of each of these sequences.
9. Enumerate the causes of childhood proptosis. Describe the characteristic CT findings in any two such conditions which may present with childhood proptosis.
10. Enumerate the causes of periosteal reaction in childhood. Discuss the radiological approach towards its diagnosis.

Paper II

1. Explain the role of imaging in the evaluation of a patient with

varicose veins. Outline the various radiological interventional approaches to the management of varicose veins.

2. Describe the characteristic radiological features which define pulmonary tuberculosis. What are the possible complications of a cavitory disease?

3. Enumerate the various left to right shunts. Discuss the characteristic radiological features which can help define them.

4. Discuss the diagnostic imaging findings in stenosis of internal carotid artery at its origin. What is the role of an interventional radiologist in the management of such a case?

5. Enumerate the predisposing conditions which can lead to aspiration pneumonia. What are its characteristic radiologic features?

6. Enumerate the predisposing factors which can lead to the respiratory distress syndrome of newborn. Describe its radiological findings and state its acute and subacute complications.

7. Discuss the role of imaging in the evaluation of a patient with acute lower limb ischemia. What is the role of an interventional radiologist in the management of such a case?

8. Enumerate the possible causes of a sudden unilateral loss of vision. Discuss the role of a radiologist in the diagnosis of any two such conditions which may present in this manner.

9. Describe the CT and MR imaging features in malignant mesothelioma. How is its staging carried out?

10. Enumerate the various types of aortic dissection. Describe its contrast enhanced computed tomography and magnetic resonance imaging findings stating their respective advantages and limitations.

Paper III

1. Enumerate the causes of acute intestinal obstruction. Describe the role of a radiologist in its diagnosis.

2. Describe the ultrasound and Doppler evaluation of a suspected case of portal hypertension. Outline the indications and procedure of trans jugular intrahepatic porto-caval shunt.

3. Define the role of imaging in the evaluation of a patient with

perianal discharge. Draw a labelled diagram of coronal section of anorectal junction region, depicting the internal and external sphincter, and portraying the types of anorectal fistulae.

4. Enumerate the types of congenital malrotation of the gut. Describe the embryological basis of their occurrence.

5. Discuss the physical principle of magnetic resonance cholangiopancreatography. Describe the various anatomical variants of the biliary tree.

6. Enumerate the possible acute complications which may occur in a renal transplant recipient. Define the role of a radiologist in their diagnosis.

7. Enumerate the causes of a unilateral adnexal mass. Discuss the role of a radiologist in diagnostic evaluation of such a patient.

8. Enumerate the causes of menorrhagia in a perimenopausal patient. Discuss the role of a radiologist in diagnostic evaluation of such a patient.

9. Enumerate the congenital foetal cardiac anomalies that can be detected on antenatal ultrasound examination. Describe the ultrasound appearance of any two such conditions.

10. a) Intrauterine image-guided interventions which can be carried out in pregnancy.

b) Role of Doppler ultrasound in the evaluation of foetal middle cerebral artery.

Paper IV

1. Draw labelled diagram(s) of the suprahyoid neck spaces.

2. A patient develops severe hypotension immediately following the intravenous administration of an iodinated water-soluble contrast medium. How would you manage such a patient?

3. Define the terms “Roentgen”, “Rad”, “Gray” and “Sievert”. State the acceptable upper limits of radiation exposure in a radiation worker as per the AERB guidelines.

4. What are the types of nephrotoxicity reactions associated with

water-soluble contrast media and what are their defining features? What are the proposed mechanisms which produce contrast induced nephrotoxicity?

5. Discuss the pre-procedure work-up, procedure, and the postprocedure care in a patient scheduled to undergo a tru cut/core biopsy for a hepatic mass lesion.

6. State the age of appearance and fusion of various epiphyses around the elbow joint on plain radiographs in a male subject. Discuss how radiography can help in gender determination in an adult subject.

7. Describe the physical principle behind various diffusion weighted imaging techniques. Discuss their clinical usefulness.

8. Define the statistical terms “accuracy”, “positive predictive value” and “negative predictive value”, citing their clinical usefulness.

9. Enumerate the constituents of ultrasound jelly. State the significance of each constituent.

10. Enumerate the MR contrast agents. Discuss their potential adverse effects.

December 2017

Paper I

1. What are causes of hypoxic Ischemic encephalopathy in children and adults? Enumerate imaging features.

2. What are the varieties of osteosarcoma? Enumerate the imaging features.

3. What are the differential diagnoses in unilateral exophthalmos in a young adult? Enumerate the radiological investigations and their role and techniques of intervention in carotico-cavernous fistula.

4. Enumerate masses of developmental origin in the neck. Discuss imaging features in each.

5. What are the clinical features in a patient with rheumatoid arthritis? Describe the various imaging characteristic in rheumatoid arthritis?

6. Enumerate the causes of presacral masses in children. Discuss the various imaging features of each.
7. Enumerate the causes of adrenal enlargement in an adult of 40 years. Discuss the imaging features of adrenal adenoma.
8. Define stress fracture. Enumerate the common sites. Discuss the role of imaging in early diagnosis.
9. Discuss the role and imaging features of MRI in multiple sclerosis.
10. Describe the arterial anatomy of Circle of Willis. Discuss the role of imaging in intracranial aneurysm.

Paper II

1. Enumerate the benign cardiac tumours and discuss their imaging features.
2. Imaging features of:
 - (a) Coarctation of Aorta.
 - (b) Total Anomalous Pulmonary Venous Connections (TAPVC).
3. Enumerate the causes of anterior mediastinal masses and discuss their imaging features.
4. Discuss the CT features of:
 - (a) Silicosis
 - (b) Pulmonary embolism
5. Discuss the causes of pneumothorax with their imaging features.
6. Enumerate the various causes of impotency. Discuss the role of Penile Doppler in the diagnosis of erectile dysfunction.
7. A 26 year male presents in Emergency with abdominal trauma.
 - (a) Enumerate the various causes.
 - (b) Discuss the role of imaging in any two organs.
8. Role of MRI in lung cancer staging.
9. Etiology and radiological finding in acute respiratory distress syndrome.
10. Role of Color Doppler in:
 - (a) Renal hypertension
 - (b) Acute limb ischemia

Paper III

1. Current indications and limitations of Foetal MRI.
2. Causes and CT features of small bowel obstruction
3. Enumerate the causes of primary infertility in male & female and discuss the role of imaging and interventions in any two causes.
4. Causes of obstructive jaundice and role of imaging in them along with therapeutic Interventions.
5. Enumerate the causes of unilateral small kidney. Discuss the role of Imaging in them.
6. What is enteroclysis? Discuss its technique with merits and demerits with Barium meal follow through.
7. Classify the renal injuries and describe briefly the relevant interventional procedures in renal injuries.
8. Enumerate the different types of Aortic Aneurysm, their sites and interventional management.
9. What is testicular torsion? Discuss briefly the different types of torsion and diagnostic Imaging features?
10. a) Embryological development of Pancreas.
b) Enumerate the causes and describe the spectrum of CT findings in acute pancreatitis.

Paper IV

1. Principle, equipment, uses and adverse effects of RFA.
2. Principle and role of MR Spectroscopy.
3. a. Sono saline hystero-graphy
b. Sono mammography
4. Principle and clinical applications of Elastography.
5. Principle and role of PET scanning in benign conditions affecting the lymph nodes
6. Role of plain X-ray of wrist in determining the age of child in medicolegal cases.
7. Name the hepatic interventions done through the transjugular route. Discuss the indication, technique and complications of transjugular liver biopsy.

8. Discuss briefly the type and structure of radiographic film and imaging film.
9. Describe arterial anatomy in brief of lower limbs and discuss the diagnostic and interventional techniques in arterial stenosis/ occlusion of lower limb arteries.
10. Define the objectives and general guiding principles and basic factors for protection against radiation.

June 2017

Paper IV

1. a) Define the PC & PNDT Act, its objectives, its target bodies and requirements.
b) Enumerate the essential information to be filled in Form F and the documents to be attached along with.
2. a) Delineate the boundaries of the middle cranial fossa.
b) Enumerate the anatomical structures found in the middle cranial fossa.
3. a) Enumerate the constituents of the ultrasound gel.
b) Identify the usefulness of each constituent.
c) Explain the principle behind the usage of ultrasound gel in sonographic imaging.
4. With reference to diagnostic imaging, explain the following terms stating their clinical significance:
a) Signal to noise ratio(SNR)
b) Shear wave elastography
5. a) Name the essential drugs that must be kept in readiness in a radiology room to tide over the contrast reactions caused by iodinated contrast agents.
b) State their usage in diverse contrast reaction related emergencies citing their dosage and mode of administration.
6. a) What is the role of nuclear medicine imaging techniques in the investigation of neuroendocrine tumours?
b) Pitfalls of these techniques.

7. a) Sensitivity, specificity, positive predictive value and negative predictive value of a diagnostic test.
b) Quote an example using these values for any radiological test.
8. Basic principles of Dual Energy Computed Tomography and its clinical applications in abdominal imaging.
9. Enumerate the current interventional radiology techniques being employed in the management of acute massive pulmonary thromboembolism. Outline the merits and limitations of each.
10. Recent developments in non-invasive clinical lymphatic imaging techniques capable of visualizing the central lymphatic anatomy and flow dynamics.

Paper III

1. a) Based on the anatomic location of the bleeding site, how is gastrointestinal bleeding typically categorized?
b) How does multi-detector computed tomographic angiography help in the localization and characterization of acute gastrointestinal bleeding?
2. Role of contrast enhanced ultrasound in the imaging of nodules in a cirrhotic liver, and the algorithm of how it can be integrated to be a part of a multimodality imaging approach.
3. a) Enumerate the venous tributaries constituting the portal venous system and the sites of portosystemic anastomosis.
b) Sonographic and CT imaging features of portal hypertension.
4. a) Multi-detector computed tomography (MDCT) features of acute intestinal ischemia.
b) How does MDCT contribute to appropriate treatment planning and provide important prognostic information?
5. CT imaging features of acute pancreatitis alongwith the revised Atlanta system of classification.
6. A 52 year old road traffic accident victim is brought to you from the casualty with history of frank haematuria.
a) How would you evaluate this patient?
b) What are the possibilities with their key radiological findings?

7. A 66 year old patient has undergone a renal transplant. Enumerate the potential complications that may occur in this patient. What is the role of Colour Doppler and ultrasound in the imaging of these complications?
8. A 26 year old patient, who is 12 weeks postpartum, is referred to you for radiological appraisal with a history of bleeding per vaginum and raised beta-HCG levels.
 - a) What is the likely diagnosis?
 - b) How would you evaluate this patient?
 - c) Mention the possibilities with their key radiological findings.
9. a) How would you decide on the amnionity and chorionity in a multiple pregnancy?
 - b) Enumerate the complications that may occur in a twin pregnancy.
 - c) Radiological findings in twin-twin transfusion syndrome.
10. a) Enumerate the various Müllerian duct anomalies along with the line diagrams.
 - b) Imaging algorithm and diagnostic features of the various types of Müllerian duct anomalies.

Paper II

1. a) Pathophysiology of pulmonary edema.
 - b) Radiographic and CT signs of pulmonary edema.
2. CT imaging features of:
 - a) Constrictive pericarditis.
 - b) Pulmonary sequestration
3. A 9 year old girl presents with acute right lower quadrant abdominal pain:
 - a) Enumerate the likely causes.
 - b) Imaging features of the two most common conditions.
4. Imaging features of:
 - a) Tetralogy of Fallot (TOF)
 - b) Ebstein's anomaly.
5. a) Technique of High Resolution Computed Tomography (HRCT) of the lung.
 - b) Radiological features of usual interstitial pneumonia.
 - c) How does it differ from non-specific interstitial pneumonia?

6. Radiological findings in:
 - a) Chance fracture
 - b) Lunate and perilunate dislocations.
7. Role of imaging in a patient with suspected splenic trauma.
8. Colour Doppler flow imaging of carotid arteries.
9. a) Indications and technique of CT coronary angiography.
- b) Imaging features of lymphangitis carcinomatosa.
10. a) Imaging algorithm in a 40 year old man with hemoptysis.
- b) Role of interventional radiology in this case.

Paper I

1. A 40 year old man on high dose steroids for interstitial lung disease presents with pain in the right hip with inability to walk:
 - a) What possibilities would you consider?
 - b) What imaging protocol would you adopt for this patient?
 - c) Enumerate the key imaging findings for the likely entity.
2. A 26 year old female presents with intractable epilepsy and is being worked up for possible surgery:
 - a) Epilepsy protocol for imaging this patient.
 - b) Radiological features of hippocampal sclerosis.
3. a) Anatomy of the osteomeatal unit, any two variants.
- b) Radiological features of fungal sinusitis on CT & MRI.
4. a) How would you evaluate a bone tumour on a plain radiograph?
- b) Radiological features of chondroid tumours on plain radiograph.
5. A mother of a 2 year old child notices 'something white' in one eye. The ophthalmologist refers the child for imaging:
 - a) What possibilities would you consider? b) What imaging protocols would you follow?
 - c) Enumerate key imaging findings for one most likely entity.
6. a) Boundaries and contents of the carotid space of the neck.
- b) Imaging findings of vagal schwannoma.
7. A 45 year old lady presents with bone pain, renal calculi, weakness and psychiatric symptoms. X-Ray hand shows subperiosteal resorption in middle phalanx:

- a) What possibilities would you consider?
 - b) How will this patient be investigated radiologically further?
 - c) Enumerate key imaging findings of the likely entity.
8. A 30 year old man who has HIV and at present has low CD4 count presents with gradual cognitive impairment and is referred for imaging:
- a) What are the likely possibilities?
 - b) Radiological findings in progressive multifocal leukoencephalopathy.
9. a) Differences between pyogenic and tubercular osteomyelitis.
b) Imaging findings of tubercular spondylitis in MRI.
10. A 54 year old man presents to the hospital with a history of sudden severe headache. There is no history of trauma or intoxication:
- a) What possibilities would you consider?
 - b) Imaging features for the most likely diagnosis in this patient.
 - c) Radiological intervention that may be required for this patient.

December 2016/April 2017

Paper I

1. A 36-year-old woman presents with progressive deafness and tinnitus in the left ear. She undergoes magnetic resonance imaging of the brain, which demonstrates a 2 cm mass at the left cerebellopontine angle. Enumerating the differential diagnosis, write the key imaging findings for each possible entity.
2. A 28-year-old man presents with a history of recent head injury and unilateral, painful, pulsatile exophthalmos with reduced visual acuity.
 - a) What possibilities would you consider?
 - b) What imaging protocol would you adopt for this patient, and why?
 - c) Enumerate the key imaging findings for the likely entity.
3. MR findings in a patient of suspected multiple sclerosis.
4. A 26-year-old man presents with pain in the jaw. An orthopantomogram demonstrates a radiolucent lesion in the mandible. Write the differential diagnosis using an algorithmic approach while citing the key imaging findings for any 2 possible conditions.

5. A 30-year-old woman with a past medical history of spontaneous pulmonary embolism presents two weeks after giving birth with severe headache, vomiting and drowsiness.
- a) What imaging protocol would you follow for this patient, and why?
 - b) What would be the key imaging findings?
6. A previously well 62-year-old woman sustains a subcapital fracture of the left neck of femur following a fall onto the hard ground. The plain film reveals multiple lytic lesions within the pelvic bones and proximal femur.
- a) How would you evaluate her as a radiologist?
 - b) Write the differential diagnosis, enumerating the key imaging features of any 2 conditions.
7. a) Enumerate the components of the rotator cuff of the shoulder.
b) Imaging findings in a patient with rotator cuff tear.
8. A 3-year-old child presents with fever, bleeding from the gums and bone pain. His wrist radiograph demonstrates lucent metaphyseal lines in the distal radius and ulna. Write the differential diagnosis, enumerating the imaging features of any two likely pathologic conditions which may present with this clinico-radiological picture.
9. Enumerate the conventional radiographic patterns of bone destruction. List differential diagnosis of each.
10. State in brief the imaging appearance and clinical significance of the following signs in musculoskeletal imaging:
- a) Fallen fragment sign
 - b) Bone within a bone sign
 - c) Shiny corner sign
 - d) Rugger jersey spine

Paper II

1. A 32-year-old man presents to his family physician with cough, fever and increasing shortness of breath. A chest radiograph reveals the presence of symmetrical hilar masses. The masses have a lobulated and well defined outline with some fine calcification. The only significant element in this man's occupational exposure history is he is fond of keeping birds as pets

in the house. Write the differential diagnosis in this clinico-radiological setting, detailing the chest CT findings in any two likely conditions.

2. Write the characteristic features of Primary and Post primary pulmonary tuberculosis on a chest radiograph, while elucidating the role of CT in determining disease activity.

3. a) Draw a labeled diagram depicting the segmental anatomy of the left lung.

b) Classic radiographic features of left upper lobe collapse on a chest radiograph.

4. Radiological features of:

a) Total anomalous pulmonary venous connection (TAPVC)

b) Hyaline membrane disease

5. A 78-year-old patient undergoes an endovascular aortic aneurysm repair (EVAR).

a) Outline the current reference standard post-EVAR surveillance imaging drill that the patient requires as a part of his routine follow-up.

b) Enumerate the potential complications with their key imaging findings which need to be recognized and appropriately reported during the follow up.

6. Enumerate the causes of acute right upper quadrant pain and imaging (USG & CT) findings in two common conditions.

7. a) Define the role of imaging in renal trauma.

b) How are the renal injuries graded? Briefly portray the imaging findings in diverse types of renal injuries.

8. a) Define a solitary pulmonary nodule.

b) Morphological and functional imaging evaluation of solitary pulmonary nodule.

9. Role of Color Doppler Flow Imaging (CDFI) in portal hypertension.

10. a) Define acute respiratory distress syndrome.

b) Write the imaging findings of acute respiratory distress syndrome on chest radiography, lung ultrasound and computed tomography and the respective role of these modalities in its diagnosis and management.

Paper III

1. A 40-year-old man is suspected to have developed a small bowel perforation.
 - a) Enumerate its possible causes.
 - b) How would you evaluate this patient?
 - c) State the specific imaging features of any two such pathologic conditions, which may produce small bowel perforation.
2. Imaging findings in chronic pancreatitis and differentiation of focal mass in chronic pancreatitis from carcinoma of pancreas.
3.
 - a) Spectrum of imaging features in primary small bowel lymphoma.
 - b) Tabulate the correlation between histopathological and radiological findings.
4. Role of current imaging techniques in the localization and characterization of acute gastrointestinal bleeding.
5. Role of imaging in the diagnosis and management of gallbladder cancer.
6.
 - a) Multiparametric MRI in evaluation and staging of prostate cancer.
 - b) Role of TRUS biopsy.
7.
 - a) Differential diagnosis for a complex cystic renal lesion.
 - b) Imaging findings in any two clinical entities which are categorized as heritable multifocal cystic renal disease.
8. A 26-year-old man presents with fever, dysuria, dull lumbar pain, microscopic haematuria and sterile pyuria. Considering the most likely diagnosis in this setting, what would be the possible findings on an intravenous pyelography examination?
9.
 - a) What do you understand by the term “pelvic inflammatory disease” (PID)?
 - b) Role of diverse imaging modalities in a suspected case of PID.
 - c) State the key imaging findings in Fitz-Hugh-Curtis Syndrome.
10.
 - a) Enumerate the types of morbidly adherent placenta along with the predisposing conditions.
 - b) Key imaging findings on ultrasound (including CDFI) and MRI.

Paper IV

1. a) What are subarachnoid cisterns?
b) What is their clinical significance during imaging?
c) Name any six subarachnoid cisterns stating their anatomical location.
2. What are the various regulatory controls exercised by the Atomic Energy Regulatory Board for setting up and maintenance of medical diagnostic radiology equipment in India?
3. a) With reference to imaging, what do you understand by the term “elastography”?
b) List the key components of MR elastography.
c) What are the relative strengths and limitations of ultrasound and MRI-based elastography techniques in staging liver fibrosis?
4. Role of radionuclide imaging in thyroid and parathyroid disorders.
5. a) Define contrast-induced-nephropathy.
b) What are its proposed mechanisms?
c) What measures would you take in a patient affected with type 2 diabetes and systemic hypertension to safeguard against contrast-induced-nephropathy?
6. a) What is “bone age”?
b) What are the clinical applications of its determination?
c) Factors which can affect the bone age of a person.
d) How reliable is bone age from the standpoint of forensic radiology?
7. a) Mammography tube - differences with conventional X-ray tube.
b) Aliasing on colour Doppler and its application.
8. Role of perfusion imaging, diffusion-weighted imaging and MR spectroscopy in the diagnosis and classification of common intraaxial brain tumors in adults.
9. a) Percutaneous trans-hepatic biliary drainage.
b) Chi Square test & its usefulness in clinical research.
10. a) Side effects of gadolinium based compounds.
b) How does the safety profile vary with compound structure?
c) What new concerns have been raised against them which might act as a deterrent in their usage even in patients with normal renal function?

June 2016

Paper I

1. a) Anatomy of cranio-vertebral junction (CVJ).
b) Enumerate various diseases affecting CVJ.
c) Imaging features of traumatic injuries in this region.
2. Various MR techniques and imaging features in diagnosis of bone marrow disorders.
3. Radiological findings in: a) Dermatomyositis
b) Lipoma arborescens
4. a) Classify AV malformation in brain.
b) Imaging features and interventions in AV malformation of brain.
5. a) Draw a diagram of cross sectional anatomy of inner ear.
b) Imaging work-up in a child presenting with sensorineural hearing loss.
6. Role of radiographs, CT and MRI in diagnosis of various spinal tumors.
7. a) Bare orbit sign.
b) Fungal infection of paranasal sinuses
8. a) Basic principle and technique of MR spectroscopy.
b) Its role in differentiating various brain tumors.
9. a) Pathophysiology of hyperparathyroidism.
b) Imaging findings in a case of primary hyperparathyroidism.
10. Role of plain radiograph, ultrasound and MRI in congenital dislocation of hip.

Paper II

1. a) Enumerate various causes of right upper lobe collapse.
b) Imaging features on plain radiographs and CT in such a case.
2. Radiological findings of coarctation of aorta on a plain radiograph, barium contrast study and DSA.
3. Role of scintigraphy in cardiac imaging with emphasis on myocardial perfusion and viability.
4. a) Classify diaphragmatic hernias.

- b) Radiological techniques to establish the diagnosis with relevant imaging findings.
- 5. Imaging features and differential diagnosis of silicosis.
- 6. a) Penile arterial flow physiology.
- b) Technique and utility of duplex sonography in evaluation of erectile dysfunction.
- 7. a) Define Doppler effect.
- b) Color Doppler and Power Doppler modes.
- c) Role of RI and PI in Doppler imaging.
- 8. a) Role of various Doppler parameters in diagnosis and management of a case of IUGR.
- b) Significance of aortic isthmus index.
- 9. a) Define pulmonary sequestration.
- b) Types of pulmonary sequestration.
- c) Imaging findings in a case of pulmonary sequestration.
- 10. a) Define truncus arteriosus.
- b) Mention its type and characteristic features of its various types.
- c) Its chest radiographic, Echocardiographic and MRI findings.

Paper III

- 1. a) Enumerate causes of hypertension in a 10-year-old male.
- b) Outline radiological approach in such a case.
- c) Role of MDCT in renal hypertension.
- 2. a) Imaging features and role of various diagnostic modalities in a case of cholangiocarcinoma.
- b) Role of radiological interventions in its management.
- 3. Role of imaging in uraemia, citing the specific role and limitation of conventional radiography, sonography, CT, MR and renal scintigraphy.
- 4. Role of plain radiography, enteroclysis, ultrasonography, CT and MRI in evaluation of small bowel obstruction.
- 5. a) Enumerate the conditions under which the revised PNDT Act, 2010 permits you to conduct prenatal diagnostic techniques.
- b) What steps would you take in clinical ultrasound practice to comply with the Act?

6. a) Define abnormal endometrial thickening.
b) Enumerate its causes and discuss their imaging features.
7. a) List various causes of female infertility.
b) Role of HSG, US and MRI in their evaluation.
8. a) List various causes of bleeding in the first trimester. b) Discuss their sonographic features.
9. a) MRI and MRS findings in prostatic carcinoma and its staging.
b) Role of TRUS biopsy.
10. a) Enumerate various ovarian tumours of stromal origin.
b) Imaging features of serous and mucinous cystadenoma, and cystadenocarcinoma.

Paper IV

1. a) Define various radiation units.
b) What are the maximum permissible doses of radiation for the general public and radiation workers?
c) Ways to reduce radiation exposure during fluoroscopy.
2. Role of radionuclide imaging in renal disorders.
3. Principle and clinical applications of diffusion tensor imaging.
4. a) Anatomy of rotator-cuff in shoulder.
b) Role of MRI in SLAP lesions.
5. Photoelectric effect and its application in diagnostic radiology.
6. Enumerate various MR contrast media. Mention briefly their mechanism of action.
7. Principle and clinical applications of ultrasound elastography.
8. a) What are the current indications and limitations of foetal MRI?
b) Technique of doing fetal MRI.
9. a) What are different components of PACS? Discuss one role of each component.
b) What are the advantages and disadvantages of PACS?
10. a) Define contrast-induced nephropathy (CIN).
b) Various risk factors associated with CIN.
c) What precautions can be taken to prevent CIN?

December 2015

Paper I

1. a) Enumerate causes of periosteal new bone formation.
b) Imaging features of acute osteomyelitis and infantile cortical hyperostosis.
2. Various radiological findings in Battered Baby Syndrome and their mimickers.
3. a) Enumerate neuroendocrine tumours of pancreas.
b) Their CT and MRI imaging features.
4. a) Enumerate various indications of orbital ultrasound.
b) Role of ultrasound and Colour Doppler in a child with white reflex.
5. Imaging features and interventions in vein of Galen aneurysmal malformation.
6. a) Enumerate the causes of epilepsy.
b) Distinguishing features of cortical lesions associated with epilepsy.
7. a) Enumerate the infrahyoid spaces. b) Imaging features of carotid body tumour.
8. a) Classify spinal dysraphism.
b) Pathology and imaging features of diastatomyelia.
9. Radiological findings in:
a) Dermatomyositis.
b) Lipoma arborescens.
10. a) Role of skeletal radiography in determination of bone age in a male likely to be between 12-18 years of age.
b) Factors affecting bone growth and remodeling

Paper II

1. a) Enumerate pulmonary manifestations in patients with HIV.
b) Chest X-ray and CT features in *Pneumocystis carinii* pneumonia.
2. a) What is truncus arteriosus?

b) Its classification, imaging features on chest X-ray, echocardiography and MRI.

3. Radiological features of:

a) McLeod's Syndrome

b) Vanishing Lung Syndrome

c) Scimitar syndrome

4. a) Enumerate the causes of SVC obstruction

b) Its pathophysiology and imaging features.

5. a) BIRADS classification.

b) Imaging features of Phyllodes tumour on mammography, ultrasound and MRI.

6. Imaging evaluation in a 13 year old girl presented with acute right lower quadrant pain.

7. a) Classify renal injuries.

b) Imaging features and interventions in them.

8. a) Pathophysiology of venous incompetence in lower extremity. b) What are common locations of Perforators?

c) Colour Doppler findings and interventions of venous incompetence in lower limb.

9. Pathophysiology, imaging features, complications and differential diagnosis of Respiratory Distress Syndrome of Newborn.

10. Role of CT and MRI in staging of lung cancer.

Paper III

1. a) Enumerate the causes of obstructive jaundice in adults.

b) Role of USG & MRI in the evaluation of obstructive jaundice.

2. a) Characteristic pathological features of gastrointestinal stromal tumours along with the imaging findings.

b) Management options and post treatment follow up of these tumours.

3. a) Enumerate the congenital anomalies affecting the uterus. b) MR Imaging findings in these.

4. a) Pathophysiology and imaging features in inflammatory bowel disease (IBD).

b) Role of CT enteroclysis in IBD.

5. a) Enumerate the various causes of nodules in a cirrhotic liver.
b) Role of Imaging in differentiating them.
6. Important differential diagnosis in a child with acute scrotum and their imaging features.
2. Imaging features of: a) Caroli disease. b) Biliary atresia
3. a) Bosniak classification of renal cysts.
b) Imaging features of renal lymphoma.
4. a) MR imaging of normal prostate: Technique & zonal anatomy.
b) Role of MR imaging in staging of carcinoma prostate.
5. a) Antenatal diagnosis of congenital diaphragmatic hernia.
b) Imaging findings in gestational trophoblastic disease.

Paper IV

1. a) What is scatter radiation? How does it affect radiographic image quality?
b) Methods to reduce scatter radiation.
2. a) Enumerate the types of digital radiography. Describe each one briefly.
b) Advantages and limitations of digital radiography compared with conventional film screen radiography.
3. a) Principles and techniques of dual energy CT.
b) Clinical applications of dual energy CT.
4. Principles and techniques of ultrasound elastography along with its clinical applications.
5. a) Define contrast induced nephropathy (CIN).
b) Conditions predisposing to CIN and the precautions to be taken to avoid CIN.
6. a) HIFU – Principles & clinical uses.
b) PET-CT in staging of brain tumours.
7. a) Techniques for dose reduction in MDCT.
b) How is mammography tube different from conventional X ray tube?
8. Indications, imaging evaluation and technique of Transjugular Intrahepatic Portosystemic Shunt(TIPS).
9. a) Documentary requirements under PCPNDT Act.
b) Positioning and technique for Water's view

10. a) Advances in lead apron technology.
- b) Technique of MR perfusion and its clinical applications in brain lesions.

System Wise questions:-

ABDOMEN

1. Differential diagnosis of mass in right iliac fossa.
2. Describe the role of x ray, CT, USG and MRI in acute abdomen.
3. Superior mesenteric artery syndrome.
4. Abdominal tuberculosis. Discuss the features that are useful to differentiate it from Crohn's disease.
5. Retroperitoneal fibrosis.
6. Describe imaging in a 5 years old child presenting with a lump in the right lumbar region.
7. Imaging in blunt abdominal trauma.
8. Prune belly syndrome.
9. Imaging of Retroperitoneum(all modalities and their pros and cons)
10. Imaging in retroperitoneal fibrosis.
11. Imaging in paediatric, adult male and female presenting with right lower quadrant pain.
12. Draw a neat line diagram of perinephric spaces including its relationship with other spaces. Write CT features of perinephric abscess and urinoma.
13. Enumerate various causes of Para vertebral masses and their imaging features
14. Enumerate the causes of mechanical small bowel obstruction in an adult. Describe the differentiating features of small and large bowel obstruction on imaging. Briefly discuss the role of CT in mechanical small bowel obstruction.
15. Acute and chronic pancreatitis.

16. Briefly discuss the development of midgut. Describe the imaging findings of midgut malformation and midgut volvulus on various imaging modalities.
17. Briefly describe the role of imaging in the following: a) Neuroendocrine tumors of pancreas b) Small bowel lymphoma.
18. Enumerate the causes of a palpable lumbar mass in a 5-yr-old child. Discuss the algorithmic approach you would use to arrive at diagnosis in this case.
19. Enumerate the causes of pneumoperitoneum with peritonitis in a 30 yr old male patient. Describe the findings which can be seen in supine abdominal radiograph in a case of pneumoperitoneum. Discuss the CT findings which may be seen in bowel ischemia due to acute superior venous thrombosis.
20. Pneumotosis intestinalis.

ADRENALS

1. CT features in adrenal tumours.
2. Pheochromocytoma.
3. Total evaluation techniques for adrenal disease.
4. Classify adrenal tumors and role of CT and MRI in evaluating them.
5. CT and MRI anatomy of Adrenal glands and normal variants.

ANATOMY

1. Anatomy of urethra.
2. Segmental anatomy of liver and its importance.
3. Cross-Sectional labeled diagram of Peritoneal Spaces at level of renal hila.
4. Cross Sectional Anatomy of SupraRenal level. Enumerate the hormones elaborated by zones of the Supra renal glands.
5. Anatomy of Maxillary sinus and classification of various pathologic diseases.
6. Radiological anatomy of Mediastinum

7. Radiological anatomy of Sella turcica and imaging features of suprasellar masses. [Jul 99/Dec 10]
8. Describe the anatomy of Gastro-oesophageal junction and imaging of hiatus hernia. [JAN 00]
9. Segmental anatomy of Lungs. [JAN 01, 02,10]
10. Orbit [DEC 02]
11. Temporal bone and Internal Auditory Canal.
12. Middle ear.
13. Neck space CT anatomy. [JUN 04]
14. CSF pathways.
15. Blood brain barrier. [DEC 05]
16. Describe normal Anatomy of Knee as seen on MRI. [09]
17. Describe the normal anatomy of coronary arteries and discuss the role of MDCT in coronary artery diseases. [09]
18. Lymphatic drainage of Lungs.
19. Embryology of Gastrointestinal Tract.
20. Embryology of Genitourinary Tract.
21. Embryology of Diaphragm.
22. Radiological anatomy of Larynx and Pharynx.
23. Anatomy and lesions of parapharyngeal spaces.
24. Radiological anatomy of duodenum and relations.
25. Radiological anatomy of pancreas.
26. Peritoneal ligaments and mesenteries (pathways of intra-abdominal disease spread).
27. Blood supply of the large intestine.
28. Radiological anatomy of carotid artery and branches.
29. Anatomy of the Biliary tree and investigations for evaluation.
30. Anatomy of the Circle of Willis with Diagram. Enumerate the causes of Subarachnoid Hemorrhage. [2010]
31. Describe the Embryology and development of pancreas. Describe the imaging features of any one important congenital anomaly of pancreas. [2010]
32. Describe anatomical variations in Circle of Willis with the help of a diagram. Enumerate the sites of intracranial aneurysm.

33. Describe the embryogenesis of the human urinary system using labelled diagrams. briefly discuss the basis of any 3 congenital defects of the kidney.
34. Describe with help of labeled diagram-vascular anatomy of testes. Explain briefly its clinical relevance in imaging of testicular malignancies.
35. Draw a neat line diagram of perinephric space including its relationship with other spaces. Write CT features of perinephric abscess and urinoma. Anatomy of cranio-vertebral junction (CVJ). b) Enumerate various diseases affecting CVJ. c) Imaging features of traumatic injuries in this region. [3+2+5 June 16]
36. a) Draw a diagram of cross sectional anatomy of inner ear. b) Imaging work-up in a child presenting with sensorineural hearing loss. [5+5 June 16]
37. Describe the anatomy of larynx with the help of diagrams. How does radiology help to assess staging of laryngeal malignancy?

BREAST

1. Breast Masses (Both benign and malignant and their imaging findings which includes xray, usg and mri))
2. Indications of X-ray mammography, Sonography and MRI of Breast.
3. Discuss the recent advances in Mammography.
4. Mammographic features of Carcinoma Breast.
5. Sonography in solid breast masses.
6. Conventional mammography techniques.
7. X-ray Mammographic tube and Breast mammographic views
8. Computer aided detection (CAD) in Mammography.
9. Screening mammography — Current status.
10. Describe BIRADS classification and approach to breast lesion with a flow chart.
11. Ultrasound Elastography in Breast lesions.

12. Briefly describe diffusion protocol for MRI breast and characterization of benign and malignant breast lesion.

STATISTICS

1. Write short notes on: A.What is p value? What is its significance and clinical applications in research? B. Sensitivity C. Specificity D. Positive and negative predictive value.
2. Define the statistical terms “accuracy”, “positive predictive value” and “negative predictive value”, citing their clinical usefulness.
3. a) Kappa measure (in terms of statistical analysis of medical data). b) The Receiver Operating Characteristics (ROC)
4. a) What is p value? What is its significance and clinical application in research? b) What is sensitivity, specificity, positive predictive value, negative predictive value and accuracy?

CARDIOVASCULAR SYSTEM

1. Doppler ultrasound versus MR angiography of carotid vessels.
2. Cardiac and pericardial calcification.
3. Role of plain skiagram chest in the diagnosis of pulmonary Hypertension.
4. Scimitar syndrome.
5. Pathogenesis of ASD.
6. Atrial myxoma.
7. Amyloid heart diseases.
8. Imaging of the extracranial carotid arteries.
9. Pathogenesis and classification of Dissecting Aneurysm of Aorta.
10. Coarctation of aorta.
11. Role of doppler study in lowest extremity arterial disease.
12. Imaging in aorto-arteritis.
13. Causes and imaging features of pericardial effusion.
14. Abdominal aortic aneurysm.

15. Role of Doppler in peripheral arterial diseases.
16. MRI in cardiac disease.
17. Tetralogy of Fallot.
18. Pulmonary Stenosis.
19. Imaging in ischaemic heart disease.
20. Aortic Dissection - Imaging and Intervention
21. Plain X-ray cardiomyopathy
22. MRI in Cardiac Imaging
23. Pathophysiology and imaging of Mitral valve disease.
24. Congenital anomalies of aortic arch and major branches.
25. Superior Vena Cava obstruction.
26. Total anomalous pulmonary venous drainage.
27. Takayasu's disease or Non-specific aortoarteritis.
28. Left to right shunts/Extracardiac Left to Right shunts.
29. Enlarged Left atrium.
30. Ebstein's anomaly.
31. Radiological approach in Cyanotic heart disease.
32. Causes and imaging features of constrictive pericarditis.
33. Enumerate congenital anomalies of IVC. Comment on the role of MR I in their diagnosis.
34. Imaging of cardiac tumors.
35. Enumerate various types of transposition of great vessels.
Describe imaging features of total anomalous pulmonary venous drainage.
36. Interventional management of deep vein thrombosis.
37. Imaging in Intermittent claudication of Lower limb.
38. Imaging in 14 years old with hypertension.
39. Imaging of PDA.
40. Assessment of correctness of positioning of various catheters and tubes as seen on chest radiographs.
41. Cardiac CT.
42. Discuss the role of MR in evaluation of pericardium and its pathologies.
43. What are the causes of pulmonary venous hypertension? Describe plain X-ray findings in pulmonary venous hypertension.

44. Radiological approach in Acyanotic heart disease.
45. Classify peripheral vascular malformation. Describe sonographic color doppler, MRI and angiographic features of venous malformation. Mention suitable embolic material for their interventional management.
46. Describe the imaging findings of Coarctation of aorta(differentials for rib notching) on plain radiograph, barium contrast study, DSA and MRI . Briefly describe the role of interventional radiology in it.
47. Describe plain radiographic findings in Rheumatic heart disease in Mitral Stenosis. Mitral regurgitation with mitral stenosis and Aortic stenosis.
48. Classify aortic dissection. Describe the role of CT angiography in diagnosis and management of aortic dissection.
49. Enumerate causes of acute chest pain in an elderly patient. Briefly describe CT findings in 3 common likely condition.
50. Classify congenital cardiac abnormalities. Briefly discuss abnormalities of Situs and Looping (or topology) with their imaging features.
51. How will you radiologically investigate a 60 year old hypertensive & diabetic female presenting with severe chest pain of acute onset? Briefly discuss imaging features of the most common cause for it. Also describe the role of radiology in its complications.
52. Enumerate the radiographic features of the enlarged right atrium and enlarged left atrium. Briefly describe lung field changes in case of mitral stenosis.. What other imaging techniques will be useful in making the diagnosis? Briefly highlight the significance of each.
53. Briefly describe the anatomy of the pericardium. List various causes and imaging findings in a case odd constrictive pericarditis.
54. Describe the arterial anatomy of the carotid vascular system with the help of labeled diagrams. Discuss the role of ultrasound and; color Doppler imaging in evaluation of extra cranial carotid occlusive diseases.
55. What is Eisenmenger Syndrome? Enumerate the conditions that may produce this syndrome. Discuss its key radiological features.

56. Enumerate the radiologically-evident pericardial afflictions on a chest radiograph. Describe their key radiological findings.
57. What are the indications of coronary CT angiography? Describe the techniques of performing coronary CT angiography. What do you understand by Calcium score and what is its clinical relevance? Describe the methods to reduce the radiation dosage to patients during performance of MDCT coronary angiography.
58. Enumerate the various causes of bilateral weak femoral arterial pulsations in a 20 year old female patient. Describe the imaging findings in any two important causes.
59. Define and enumerate causes of restrictive cardiac diseases. Discuss the role of various imaging modalities along with imaging features in two such diseases.
60. a) Doppler assessment of AV fistula of hemodialysis access. b) Role of MDCT in cyanotic heart disease
61. a) Vascular compression syndromes in abdomen and pelvis
62. a) Role of different imaging modalities in evaluation of a case of limb ischemia. b) Role of interventional procedures in these patients.
63. Imaging features and interventions in the vein of Galen aneurysmal malformation..
64. a) What is truncus arteriosus? b) Its classification, imaging features on chest X-ray echocardiography and MRI.
65. a) Enumerate the causes of SVC obstruction. b) Its pathophysiology and imaging features.
66. a) Role of ultrasound in the evaluation of a suspected vascular malformation. b) Role of interventional radiology in low flow vascular malformations.
67. Enumerate the congenital foetal cardiac anomalies that can be detected on antenatal ultrasound examination. Describe the ultrasound appearance of any two such conditions.
68. a) Myocardial viability imaging.
69. What are the indications for cardiac evaluation of “delayed hyper-enhancement” in MRI? Which MRI sequences are usually needed for DHE assessment? Describe the cardiac MRI features of Hypertrophic Obstructive Cardiomyopathy.

CHEST

1. Describe briefly the pathophysiology of Pulmonary Embolism. Give in detail the imaging modalities for diagnosis of this entity and their relative merits and demerits.
2. Wegner's granulomatosis
3. Pathogenesis and imaging of pulmonary sequestration.
4. Role of imaging in bronchogenic carcinoma(Xray, CT, MRI).
5. Discuss in brief the differential diagnosis of mediastinal masses and their radiological appearances. State the radiological basis of differentiating a mediastinal mass from an intrapulmonary mass. How would you localize the compartment of a mediastinal lesion? Draw a line diagram to define the radiological anatomy of mediastinum.
6. Alveolar Proteinosis.
7. Pleural tumours.
8. Diagnosis of pulmonary infarction.
9. Adult Respiratory Distress Syndrome (ARDS)
10. Sarcoidosis.
11. Pulmonary plethora and its distinctive features.
12. What are clinical applications of CT in evaluation of non-neoplastic lung diseases?
13. Pan-acinar Emphysema.
14. Tracheoesophageal fistula.
15. Evaluation and DD of Hilar Mass.
16. Metastatic tumors of Lung.
17. Silicosis.
18. Bronchopulmonary Aspergillosis.
19. Ground glass opacity HRCT- Significance and DD.
20. Unilateral opaque and Hyperlucent hemithorax.
21. Imaging in acute chest trauma.
22. Atypical Pneumonia.
23. HRCT in ILD .
24. Pulmonary lesions in AIDS .

25. Eventration of diaphragm .
26. Raised left Dome of Diaphragm.
27. Radiological feat. in Congenital Cystic Adenomatoid Malformation of the lung.
28. Azygos lobe.
29. What are the causes of pulmonary venous hypertension?
Describe plain X-ray findings in pulmonary venous hypertension.
30. Discuss in detail imaging features of thoracic lymphoma.
31. HRCT in Diffuse lung disease.
32. HRCT in occupational lung diseases.
33. HRCT in pulmonary tuberculosis.
34. Role of chest radiography in emergency situations.
35. DD and imaging features of para-vertebral shadow.
36. Describe etiopathogenesis, common causes, plain film and CT features of lymphangitis carcinomatosa.
37. Describe plain radiographic and CT findings of rt. upper lobe pulmonary collapse.
38. Enumerate various germ cell tumors of mediastinum. Discuss their imaging features.
39. Give in detail and management of aortic dissection.
40. Enumerate various diseases caused by inhalation of inorganic dust. Briefly describe chest radiographic and CT findings of two most common such diseases.
41. How do pulmonary arteriovenous malformation present clinically?
Discuss their chest radiographic, CT chest and angiographic findings.
Briefly mention role of interventional radiology in their treatment.
42. A 25 year old male presented with life threatening haemoptysis.
Draw an algorithm to outline management of such a case. Discuss in brief the role of chest radiograph, CT scan (with newer advances) and role of interventional radiology.
43. Enumerate the causes of superior vena cava syndrome in an adult. Briefly describe the role and findings of various imaging modalities in a case of central bronchogenic carcinoma.

44. Define pulmonary edema. What is its pathophysiology? Enumerate its causes. Describe the plain radiographic findings in pulmonary edema.
45. Write imaging findings of the following: a) Bronchial Carcinoid b) BOOP c) McLeod's Syndrome.
46. What do you understand by the term 'extramedullary hematopoiesis'? Enumerate its causes. Discuss its plain film and cross sectional imaging findings.
47. A 65 year-old chronic smoker presents with hemoptysis. The chest radiograph shows a well-defined cavitating intrapulmonary mass with spiculated margins in the left upper zone. How would you further evaluate this patient and determine the extent of disease? What would be the signs you would look for to decide if the lesion is operable?
48. Describe the changes on a chest radiograph in collapse of different lobes in both lungs.
49. Radiological findings in: a) Sequestration of lung b) Pulmonary hypertrophic osteoarthropathy.
50. Define SPN. Enumerate its causes. Discuss the radiological work up of a solitary nodule highlighting the features which enable to differentiate b/w benign and malignant nodules.
51. Name the anatomical structures which contribute to the hilar shadow seen on a frontal chest radiograph. Enumerate the causes of unilateral large hilum in a 50 yr old male. Describe the imaging findings in any 2 pathological causes.
52. Describe in brief features which are useful in differentiating Hodgkin's disease and non-Hodgkin's lymphoma.
53. Describe the radiological anatomy of diaphragm. Enumerate various types of diaphragmatic hernias. Discuss the imaging findings in any two hernias which can be seen in a 40 yr old patient.
54. a) Castleman's disease
55. Discuss various chest complications in a post-operative patient. Describe in detail imaging features in any two conditions.
56. Discuss various types of aortic aneurysms. Describe various modalities to investigate such patients with advantages and

disadvantages of each. Discuss briefly role of interventional procedure.

57. a) Takayasu's arteritis b) Role of RFA in chest tumors.
58. An adult male presents with recurrent chest infections and a cavitating lung lesion in left lower zone in a chest radiograph. Discuss the differential diagnosis and imaging features in two most likely causes.
59. Causes of pleural masses and their imaging features.
60. a) Enumerate pulmonary manifestations in patients with HIV. b) Chest X-ray and CT features in Pneumocystis carinii pneumonia.
61. Radiological features of: a) McLeod's Syndrome b) Vanishing Lung Syndrome c) Scimitar syndrome.
62. a) Enumerate etiologies of diffuse cystic lesions of the lung. b) Radiological findings in any two of them.
63. Discuss the causes of pneumothorax with their imaging features.
64. Enumerate the predisposing conditions which can lead to aspiration pneumonia. What are its characteristic radiologic features? [
65. Enumerate the various lines and tubes encountered on chest radiographs in patients admitted to the intensive care unit (ICU). Discuss the imaging findings in pulmonary oedema.
66. Enumerate the benign pulmonary tumours and discuss their differential diagnosis.
67. Outline the following using an algorithmic sequence: a) Imaging approach in a patient with blunt trauma to chest. b) Imaging approach in a patient with blunt trauma to abdomen.

CONTRAST MEDIA

1. Discuss about various MR contrast media and their mechanism of action.
2. Low osmolar contrast media.
3. Adverse drug reactions caused by I.V. Contrast media.

4. Classify idiosyncratic reactions resulting from contrast media administration. Describe the management of life threatening adverse reactions.
5. Non-ionic contrast media.
6. Management of adverse contrast reactions.
7. Recent contrast media used in USG.
8. Role of Ultrasound Contrast Agents in gastro-intestinal diseases.
9. Contrast induced nephropathy.
10. MR contrast media in Hepato biliary system/MR contrast agents for Hepatic Imaging.
11. Emergency drugs with doses that should be available in radiology department.
12. Adverse reactions of MR contrast media.
13. Nephrogenic systemic fibrosis.
14. a) What is the principle of MR contrast enhancement. b) Describe any two organ specific contrast agents and their clinical applications.
15. a) Define contrast induced nephropathy (CIN). b) Conditions predisposing to CIN and the precautions to be taken to avoid CIN.
16. What is Nano diamond?

ENT

1. Laryngeal carcinoma.
2. Imaging of temporal bone/Petrous bone.
3. Describe the imaging features of juvenile nasopharyngeal angiofibroma. Discuss the role of radiological intervention in its management.
4. Enumerate the infrahyoid neck spaces. Discuss the imaging features of pathologies of the carotid space.
5. How would you evaluate a patient of hyperparathyroidism on imaging? Enumerate the findings on plain films, CT and Scintigraphy.

6. a) Draw a diagram of cross sectional anatomy of the inner ear. b) Imaging workup in a child presenting with congenital sensorineural hearing loss.
7. Enumerate masses of developmental origin in neck. Discuss imaging features in each.
8. Computed tomographic and magnetic resonance imaging findings in mycotic sinusitis. b) Imaging features of orbital involvement in Neurofibromatosis type 1

GASTROINTESTINAL SYSTEM

1. Radiologic features of gastric malignancies.
2. Describe in brief the pathology, role of imaging and radiological features in GI tract lymphomas.
3. Necrotising enterocolitis.
4. Carcinoid tumours.
5. Radiological profile of ulcerative colitis.
6. Imaging in a Vomiting infant.
7. Intervention in upper GI bleeding.
8. Imaging in postoperative stomach.
9. Anorectal Malformations.
10. Intussusception.
11. Non-tubular inflammatory bowel disease.
12. Critical appraisal on role of small bowel enema, CT & MRI enteroclysis.
13. Colonic strictures – etiology and role of imaging in diagnosis of structures.
14. Role of CT in Epiploic Appendicitis.
15. Internal Hernias.
16. Imaging of Acute Appendicitis.
17. Enumerate causes of lower gastrointestinal bleeding. Mention current imaging techniques in their evaluation Describe the role of MDCT in its evaluation.
18. Imaging features of small bowel abnormalities in newborn.

19. CT vs MR enteroclysis for assessment of small bowel diseases.
20. Describe imaging of low intestinal obstruction in a neonate.
21. CT and Endoscopic ultrasound staging of Esophageal carcinoma.
22. Discuss various causes and imaging features in stricture of the lower end of esophagus.
23. Describe imaging features in a case of intestinal perforation.
24. Discuss the role of plain radiograph , barium studies , USG and CT abdomen in diagnosis of gastrointestinal TB.
25. Enumerate causes of mesenteric ischaemia. Briefly discuss plain radiographic, USG, CT findings and the role of intervention in this condition.
26. Name the various motility disorders of esophagus. Discuss pathophysiology and imaging features of cardiac imaging.
27. Enumerate causes of multiple nodular filling defects in small bowel. Discuss the imaging features of small bowel lymphoma.
28. Name the diseases associated with H. pylori infection. Briefly discuss barium meal features of benign and malignant gastric ulcer supported by suitable diagrams.
29. Enumerate various infections and neoplasms affecting gastrointestinal tract in AIDS. Briefly describe barium meal follow through and CT features of AIDS lymphoma.
30. Classify polypoidal lesions of the colon. Mention radiological differences between benign and malignant polyps. Discuss salient imaging features of various types of adenomatous polyps.
31. Enumerate causes of malabsorption syndrome. Describe imaging features in tropical sprue. Briefly discuss its complications.
32. Describe normal gastroesophageal junction with the help of a suitable diagram. Label various rings and lines visualized on double contrast barium swallow. Discuss imaging features of Schatzki's ring.

33. Enumerate the normal and abnormal extrinsic impressions on the cervical & thoracic parts of the esophagus during Barium swallow examination. Discuss the possibilities in a 56-year-old woman presenting with dysphagia. Describe briefly the key radiological findings in any 3 conditions.
34. Enumerate various imaging techniques employed for radiological evaluation of small bowel pathologies. Discuss the merits and demerits of each technique. Discuss in brief, CT findings in a case of ileocecal T.B. [2+5+3 June 14]
35. Enumerate various conditions associated with polypoidal lesions in the large bowel. How will you distinguish b/w benign and malignant polyps on imaging? Discuss the merits and demerits of virtual CT colonoscopy in a case of suspected familial polyposis coli. [2+4+4 June 14]
36. Technique to evaluate the stomach and imaging features of stomach malignancies.[June 15]
37. Pathophysiology and imaging features in small bowel lymphoma. [10 June 15]
38. a) Characteristic pathological features of gastrointestinal stromal tumors along with the imaging findings. b) Management options and post treatment follow up of these tumors. [(3+3)+(2+2) Dec 15].
39. a) Pathophysiology and imaging features in inflammatory bowel disease (IBD). b) Role of CT enteroclysis in IBD. [(3+4)+3 Dec 15]
40. a) Differential diagnosis of multiple colonic polyps. b) Role of double contrast barium enema in its evaluation. [4+6 Apr 16]
41. a) Radiological differences between gastric carcinoma and lymphoma. b) Role of imaging in the staging of gastric carcinoma. [5+5 Apr 16]
42. Causes and CT features of small bowel obstruction [4+6 Dec 17]
43. Define the role of imaging in the evaluation of a patient with perianal discharge. Draw a labelled diagram of coronal section of anorectal junction region, depicting the internal

and external sphincter, and portraying the types of anorectal fistulae. [6+4 Jun 18]

44. Describe the role of conventional radiography and computed tomography in the evaluation of a suspected case of acute large bowel obstruction. [5+5 Dec 18]
45. a) Imaging in esophageal motility disorders. b) Imaging in post-operative stomach. [5+5 Dec 18]
46. Enumerate the various causes of strictures in the small intestine. Describe the imaging findings in Crohn's disease of the small intestine. [5+5 Dec 18]
47. What are the causes of an acute abdomen in a neonate? What are imaging features of necrotizing enterocolitis? [4+6 Jun 19]
48. What are the sites of physiological indentation of the esophagus? Discuss radiological diagnosis of achalasia cardia and its complications. What is the role of radiological intervention? [2+5+3 Jun 19]
49. a) Name the diseases associated with H.pylori infection. b) Briefly discuss Barium meal features of benign and malignant gastric ulcers supported by a suitable diagram. [4+6 Dec 19]
Repeat from Jun 12]
50. a) Enumerate causes of Gastroduodenal artery bleed. b) Mention current imaging techniques for its diagnosis. c) Describe Interventional Radiology techniques used in its management. [3+3+4 Dec 19]

GENITOURINARY SYSTEM

1. Differentiation of Renal Cyst and Renal tumour by I.V.P. [JAN 97]
2. Diagnosis of Urinary Bladder's Tumours. [97,02]
3. Discuss the role of imaging in Renal trauma.
4. Polycystic disease of kidneys. [JAN 97, JUN 04]
5. Posterior urethral valves. [JUL 97, JUN 06]
6. Epispadias extrophy complex.
7. Pathology of renal neoplasms in the paediatric age group.
8. Acute scrotum [JUL 99, 02]

9. Cystic diseases of the kidney.
10. Child with UTI. Provide a protocol for imaging and mention their features. [JAN 00]
11. How will you investigate a case of painless hematuria? What is role of Radiologist in its management?
12. Imaging of prostate. [JAN 00, JUN 04]
13. Prostatic tumors. [JUN 03]
14. Testicular germ cell tumors. [JAN 01]
15. Radiological diagnosis of congenital lesions of kidney.
16. Interventions in upper urinary tract obstruction. [01]
17. Imaging in renal malignancies. [04]
18. Role of Doppler in testicular tumor. [DEC 04]
19. Vesicoureteric reflux. [DEC 04/09]
20. Nephrocalcinosis. [02,04,06]
21. Imaging in renal malignancies.
22. Renal tuberculosis. [Dec 05/07/09, Jun 07, Dec 14]
23. Ureterocoele. [02, 05]
24. Discuss the pathology of renal hypertension & radiological investigations for the same. [Jul 98]
25. Discuss the pathophysiology of renovascular HTN and role of imaging. [02]
26. Diagnosis of renal hypertension – present day approach . [JUN 05]
27. Imaging in evaluation of renovascular hypertension in a ten year old male. [09]
28. Discuss the role of various imaging modalities in a suspected case of reno vascular hypertension. [Jun 2008]
29. Imaging of unilateral scrotal swelling.
30. Diagnosis of non malignant prostatic enlargement .
31. Angiomyolipoma of the kidney . [Jun 06]
32. Neurogenic bladder .
33. Unilateral large kidney in a child .
34. Renal Cell Carcinoma. [Jun 04]
35. Emphysematous pyelonephritis.
36. Doppler evaluation in male impotence.

37. Classify cystic diseases of kidney and discuss role of ultrasound in these lesions. [Jun 07]
38. Enumerate causes of unilateral small kidney, Describe the role of imaging in its diagnosis. [Dec 09]
39. Mention ultrasound and Doppler findings in varicocele. Describe the role of intervention in its management. [09]
40. Enumerate the causes of Varicocele. Write US technique and US and color Doppler features in Varicocele. [June 2008]
41. MR staging of prostate carcinoma. [09]
42. Penile doppler. [06]
43. Discuss imaging of erectile dysfunction. [09, 10]
44. Color Doppler evaluation of erectile dysfunction. [Jun 08]
45. Imaging in Transplant kidney. [02]
46. Enumerate various investigative modalities for the transplanted kidney & give the normal findings in each of them. [JUL 99]
47. Doppler in renal transplant [DEC 02/09, JUN 04]
48. Role of color Doppler and ultrasound in post renal transplant patient. [09]
49. How would you evaluate donor kidney for renal transplant. Discuss role of US and Scintigraphy in various types of renal graft dysfunction. [June 2008]
50. Imaging in Non tubercular renal infections [December 2008]
51. Non vascular Interventions in upper urinary tract. [09]
52. Embryology, clinical significance and imaging of undescended Testis. [10]
53. Indications, imaging features and limitations of imaging in erectile dysfunction. [10]
54. Describe the blood supply of testes with the help of diagram. Enumerate various types of testicular torsions. Describe imaging findings in each. [Dec 10]
55. Discuss the etiopathogenesis and radiological features of renal tuberculosis. [Dec 10]
56. Enumerate the indications and describe the techniques of color doppler in Renovascular hypertension. [Dec 2010]

57. Enumerate various complications of Renal transplant and discuss their imaging findings. [Dec 2010]
58. Enumerate causes of hypertension in 10 yr old male child . Outline radiological approach in such a case. Role of MDCT and intervention in renal hypertension. [Jun 11]
59. What is vesico-ureteric reflux. Discuss its causes and grading . Briefly describe role of imaging in this condition. [Jun 11]
60. Enumerate causes of unilateral small Kidney. Discuss role of imaging in establishing the diagnosis. [June 2011] [[3+7 Dec 17]
61. Discuss the role of imaging in uraemia, citing the specific role and limitations of conventional radiography, US, CT, MRI and renal scintigraphy. [June 2011]
62. Discuss the grading of renal trauma. Describe the role of imaging in its evaluation. [4+6 Dec 11]
63. Briefly describe the penile arterial flow physiology. Discuss the technique and utility of duplex sonography in evaluation of erectile dysfunction. [3+4+3 Dec 11]
64. Describe venous drainage of testis. Discuss imaging features and interventions in varicocele. [3+4+3 Dec 11]
65. Enumerate various vascular complications in renal transplant. Briefly discuss the role of color doppler, CT, MRI and intervention in these conditions. [1+3+2+2+2 Dec 11]
66. Briefly describe MRI and MRS findings in prostatic carcinoma and its staging. Discuss role of TRUS biopsy. [4+4+2 Dec 11]
67. What are common causes of medially placed ureters? Discuss various associations. IVU, CT and MRI findings of retroperitoneal fibrosis. [2+2+2+2+2 Jun 12]
68. Enumerate causes of urethral strictures. Briefly discuss role ascending urethrogram in strictures due to trauma. Name common complications of urethral strictures. [3+5+2 Jun12]
69. Enumerate various ovarian tumors of stromal origin. Briefly discuss imaging features of serous and mucinous cystadenocarcinoma and cystadenoma. [3+2+2+3 Jun 12]

70. Describe technique of TRUS guided biopsy of prostate. Briefly mention role of contrast imaging in investigation & biopsy of a prostatic lesion. [8+2 Jun 12]
71. Enumerate the causes of hematuria in a 50 yr old male patient. Briefly discuss the role of various imaging modalities in diagnosis and staging of renal cell carcinoma. [2+4+4 Dec 12]
72. Discuss the role of imaging in uremia, citing the specific role and limitation of conventional radiography, sonography, CT, MR and renal scintigraphy. [5+2 Dec 12]
73. List the anatomical sites which may become afflicted in renal TB. Discuss their radiological features in brief. [2+8 Jun 13]
74. A 24 year old with presenting features of low grade fever, lassitude and aseptic pyuria is referred to you for radiological work up. What is the likely diagnosis?. How would you investigate this patient?. Describe the key radiological signs, specifying the changes in early, intermediate and late stages of the disease. [1+3+6 Dec 13]
75. A 38 year old RTA victim is brought to you from the casualty with history of frank hematuria. How would you evaluate this patient?. Discuss the possibilities with their key radiological findings. [4+6 Dec 13]
76. Describe penile circulation. What are the causes of male impotence? Discuss the role of Color doppler imaging in impotence. [4+2+4 Dec 13]
77. A 40-yr-old-female pt. presented with complaints of vague right lumbar pain. An USG revealed a cystic lesion in right kidney. She was advised to undergo CT scan by the radiologist for further evaluation. Enumerate possible causes. Discuss the CT protocol & findings in various lesions. [2+2+6 June 14]
78. What are the causes of painless hematuria in a 50 yr old male patient? Discuss the imaging features and role of interventional radiology in two such pathologies. [2+4+4 Dec 14]
79. a) Renal tuberculosis. [5 Dec 05/07/09, Jun 07, Dec 14]

80. A young adult male presents with painless testicular mass. What is the differential diagnosis and imaging features in the most common cause? [June 15]
81. a) Pathogenesis and imaging features of xanthogranulomatous pyelonephritis. b) Imaging features of abdominal lymphangioma. [5+5 June 15]
82. a) Classify renal injuries. b) Imaging features and interventions in them. [2+(4+4) Dec 15], [4+6 Dec 17].
83. Important differential diagnosis in a child with acute scrotum and their imaging features. [2+8 Dec 15]
84. a) Bosniak classification of renal cysts. b) Imaging features of renal lymphoma. [5+5 Dec 15]
85. MR imaging of normal prostate: Technique, zonal anatomy. b) Role of MR imaging in staging of carcinoma prostate. [3+3+4 Dec 15]
86. Role of imaging in a post renal transplant patient. [10 Apr 16]
87. Role of imaging in a young male presenting with acute scrotal pain. [10 Apr 16]
88. Imaging manifestations of renal lymphoma and its differential diagnosis. [6+4 Apr 16]
89. Enumerate the various causes of impotency. Discuss the role of Penile Doppler in the diagnosis of erectile dysfunction. [4+6 Dec 17]
90. Role of Color Doppler in: (a) Renal hypertension (b) Acute limb ischemia. [5+5 Dec 17]
91. Enumerate the causes of primary infertility in male and female and discuss the role of imaging and interventions in any two causes. [5+5 Dec 17]
92. What is testicular torsion? Discuss briefly the different types of torsion and diagnostic Imaging features? [2+(2+6) Dec 17]
93. Enumerate the causes of acute scrotal pain. Discuss the specific sonographic findings which will come in useful in making the diagnosis. [3+7 Jun 18]

94. Enumerate the possible acute complications which may occur in a renal transplant recipient. Define the role of a radiologist in their diagnosis. [2+8 Jun 18]
95. Pathogenesis of genito-urinary tuberculosis and its imaging findings in CT scan. [3+7 Jun 19]
96. Mention the various causes of infertility in male. Describe the role of imaging in the evaluation of erectile dysfunction. [3+7 Jun 19]
97. a) What are the causes of vesico-ureteric reflux? b) Describe the imaging in vesico-ureteric reflux along with its grade with well labelled diagrams [5+5 Dec 19]
98. a) Enumerate causes of painless hematuria in a 55-year-old patient. b) Discuss the imaging features and role of Interventional Radiology in any two pathologies. [4+3+3 Dec 19, *Repeat from Dec 14*]
99. a) Describe penile circulation. b) What are the causes of male impotence? c) Role of Color Doppler imaging in impotence. [3+3+4 Dec 19, *Repeat from Dec 13*]

HEPATOBIILIARY SYSTEM

1. Discuss the etiology, Classification, imaging features and complication of choledochal cyst.
2. Budd-chiari syndrome.
3. Ultrasonography features in cirrhosis liver with portal hypertension.
4. Role of imaging in obstructive jaundice.
5. Hepatobiliary intervention in Obstructive jaundice.
6. Discuss Portal Hypertension: its radiological diagnosis and interventional therapy.
7. Neonatal jaundice. [02]
8. MRCP in obstructive jaundice . [DEC 02]
9. Cystic lesions of liver . [DEC 04]
10. Benign lesions of the liver. [DEC 05, JUN 06]
11. Therapeutic interventions in liver tumors. [JUN 05]

12. Triple phase Portography. [JUN 06]
13. Doppler in hepatic cirrhosis. [DEC 07]
14. Colour Doppler and CT features in portal hypertension. [09]
15. Interventions in Hepatic tumors. [DEC 06]
16. Interventional management of Hepatocellular carcinoma. [09]
17. MRI features of hepatic hemangioma. Briefly discuss role of radiology in treatment of hepatic haemangioma. [09]
18. Radio frequency ablation of hepatic neoplasm. [09]
19. Describe CT features of liver trauma and discuss role of intervention in this. [June 2008]
20. Enumerate the causes of obstructive Jaundice. Describe technique of MRCP and its role in obstructive Jaundice. [Dec 2010]
21. What is the role of diagnostic imaging modalities in Cholangiocarcinoma. Discuss the morphological findings and the significance of various modalities in management of the disease. [Jun 11]
22. Enumerate the causes of SOL in liver . Describe the USG features in any 3 of them. [Jun 11]
23. Mention the various interventional techniques used in HCC. Briefly discuss indications and technique of two commonly employed techniques. Outline protocol for follow up in a case of HCC. [1+8+1 Dec 11]
24. Enumerate the most common cause of a 6 year old male presenting with hepatomegaly, ascites and features of portal hypertension. Discuss imaging modalities employed to investigate such patients along with various imaging features. Briefly mention role of interventional radiology in its management. [1+7+2 Jun 12]
25. A 15 day old infant has presented with prolonged conjugated hyperbilirubinemia accompanied by non pigmented stools. Name the possible etiology. Describe imaging features and various associations that may be seen in such a case. [1+6+3 Jun 12]

26. Enumerate benign hepatic masses. Describe imaging features (USG, CT, & MRI) of two commonly encountered such lesions. [2+4+4 Jun 12]
27. Describe the segmental anatomy of liver in cross sectional imaging. Discuss the role of triple phase CT in differentiating focal lesions in cirrhotic liver. [4+6 Dec 12]
28. Enumerate the common causes of obstructive jaundice. Discuss the role of various imaging modalities in its diagnosis. [2+8 Jun 13]
29. Describe etiopathogenesis of biliary atresia. Discuss the role of ultrasound, MRI and scintigraphy in assessment of biliary atresia. [2+3+2+3 Dec 13]
30. Describe pre-transplant imaging in a liver donor. What are common complications after liver transplant? Discuss the role of intervention in treating complications. [3+2+5 Dec 13]
31. What are various interventional techniques available to treat hepatic malignancies? Discuss the role of chemoembolization and radioembolization in hepatic malignant lesion. [3+4+3 Dec 13]
32. Describe the anatomy of portal venous system. What are the causes of portal hypertension? Describe the role of intervention in portal hypertension. [4+2+4 Dec 13]
33. Enumerate the causes of arterial phase enhancing focal lesions in the liver. Discuss the role of MDCT and MRI in DD of these lesions. [2+4+4 June 14]
34. a) Von Meyenburg complex. b) Imaging features of fibrolamellar HCC. [5+5 Dec 14].
35. US examination of a cirrhotic patient shows a solitary nodule in right lobe of liver. How would you investigate such a patient? Discuss in detail the role of CT, MRI and interventional radiology in such a case. [2+3+3+2 Dec 14]
36. Imaging techniques and findings in a case of Budd Chiari synd. [June 15]

37. a) Enumerate the causes of obstructive jaundice in adults. b) Role of USG and MRI in the evaluation of obstructive jaundice. [2+(4+4) Dec 15]
38. a) Enumerate the various causes of nodules in a cirrhotic liver. b) Role of Imaging in differentiating them. [3+7 Dec 15]
39. Imaging features of: a) Carolis disease b) Biliary atresia [5+5 Dec 15].
40. Indications, imaging evaluation and technique of Transjugular Intrahepatic Portosystemic Shunt (TIPS). [2+4+4 Dec 15]
41. a) Segmental anatomy of liver-Labeled diagram. b) Role of MRI in evaluation of a nodule in cirrhotic liver. [3+7 Apr 16]
42. Causes of obstructive jaundice and role of imaging in them along with therapeutic Interventions. [3+5+2 Dec 17]
43. Describe the ultrasound and Doppler evaluation of a suspected case of portal hypertension. Outline the indications and procedure of trans jugular intrahepatic porto-caval shunt. [5+5 Jun 18]
44. Discuss the physical principle of magnetic resonance cholangiopancreatography. Describe the various anatomical variants of the biliary tree. [3+7 Jun 18]
45. a) Role of magnetic resonance imaging in the evaluation of hemochromatosis and iron overload states. b) Role of sonographic and Doppler evaluation in a patient of portal hypertension. [5+5 Dec 18]
46. Describe the technique of multiphasic CT scan study of liver. Discuss the imaging findings of hepatoma. [5+5 Dec 18]
47. Tabulate the grading of liver trauma. What are the important delayed complications of injury to the liver? [5+5 Jun 19]
48. Mention causes of solitary hepatic space occupying lesions. How can MRI help in differentiating them? [4+6 Jun 19]
49. Draw a diagram of the hepatic segments. Describe the role of radiology in the work-up of a liver transplant. [5+5 Jun 19]
50. a) Mention the various interventional techniques used in the management of hepatocellular carcinoma (HCC). b) Briefly

describe the guidelines outlining the criteria of response assessment of hepatocellular carcinoma. [4+6 Dec 19, Repeat from Dec 11]

INTERVENTIONAL RADIOLOGY

1. Name the hepatic interventions done through transjugular route.
Discuss the indication, technique and complications of transjugular liver biopsy.
2. Describe arterial anatomy in brief of lower limbs and discuss the diagnostic and interventional techniques in arterial stenosis/occlusion of lower limb arteries.

MUSCULOSKELETAL SYSTEM

1. Thoracic skeletal changes associated with cardio-vascular diseases. [JAN 97]
2. Radiological features in nutritional rickets.
3. Differential diagnosis of expanding lesions of mandible. [JAN 97, JAN 01]
4. Pyknodysostosis.
5. Radiological features of spinal tuberculosis. [JUL 97, JUL 98]
6. Radiological features of congenital syphilis.
7. Neurophatic joints.
8. Radiological features of Osteosarcoma. [98]
9. DD of generalized decrease in Bone Density. [98]
10. Differential diagnosis of expanding lesions in metaphysis of long bones. [JUL 98]
11. Cleido-cranial dysostosis.
12. Pancoast tumour. [JUL 98, DEC 02,03]
13. Hypertrophic pulmonary osteoarthropathy. [JUL 98, JUL 99]
14. Enumerate the causes of Osteoporosis and use of CT in Bone Mineral Studies. [JUL 98]
15. Differential diagnosis of metaphyseal lucent lesions. [JUL 99]
16. Renal osteodystrophy.

17. Pathophysiology of renal rickets. [JAN 00]
18. Psoriatic arthritis.
19. Pathophysiology of Hyperparathyroidism. [02]
20. Radio-diagnosis of hyperparathyroidism. [JAN 00, DEC 03, JUN 04/09]
21. Radiology of Rheumatoid disease. [JAN 01]
22. Cystic jaw lesions . [DEC 02, DEC 03]
23. Perthe's disease. [02]
24. Expansile lytic lesion at upper end of Tibia. [02]
25. Hand: an index of the disease. [DEC 02, DEC 03]
26. Neurofibromatosis. OR Osseous spectrum in neurofibromatosis. [DEC 02/07/09, JUN 04]
27. Role of Skeletal Radiography in estimation of age. [02]
28. Bone age estimation [DEC 03]
29. Osteogenesis imperfecta .
30. Solitary dense vertebra. [02]
31. Imaging of Low Back pain. [02]
32. Role of MRI in bone tumors . [DEC 04]
33. Radiology of RADIO-VASCULAR SYSTEM soft tissues . [DEC 05, JUN 06]
34. Secondary hyperparathyroidism. [JUN 05]
35. Osseous lymphoma.
36. Plain film features of Acromegaly. [02]
37. Techniques for evaluation of Acromegaly . [JUN 06]
38. Basilar invagination . [JUN 05/06]
39. Radiology of a Limping Child. [DEC 06]
40. Imaging features in Mucopolysaccharidosis. [DEC 06, JUN 07]
41. Periosteal Reactions (DD). [DEC 02, 06]
42. Imaging in Tuberous Sclerosis and its associations.
43. Sickle cell disease – radiological appearances. [JUN/DEC 07]
44. Discuss causes of diffuse skeletal sclerosis and role of imaging in it.
45. Sero-negative Spondyloarthropathy. [DEC 07]
46. Differential diagnosis of radiological appearance of absorption of terminal phalanges.

47. **MR Imaging of Traumatic knee. [DEC 06]**
48. **Imaging in Meniscal tear of knee. [09]**
49. **Role of plain X rays and USG in Congenital Dislocation of Hip jt. [02]**
50. **MRI in congenital dislocation of hip joint. [09]]**
51. **MRI in SLAP lesions of shoulder. [09]**
52. **Rib Notching. [JUN 03]**
53. **Enumerate various causes of Para vertebral masses and their imaging features. [June 2008]**
54. **Mention causes of inferior RIB notching. Discuss imaging features of 2 common causes. [09]**
55. **Describe ossification of bones of elbow. [09]**
56. **Imaging and associations of Fibrous dysplasia. [09]**
57. **Briefly discuss imaging of pre-sacral masses in children. [09]**
58. **Ozone therapy for backache. [09]**
59. **Imaging findings in Plasma Cell Tumors. [09]**
60. **Evaluation of Skeletal Dysplasias in utero. [09]**
61. **Imaging in Rotator cuff lesions. [09]**
62. **What are round cell tumors of bone? Discuss in detail differentiating imaging features in these. [Jun 08]**
63. **Describe radiological features, complications and differential diagnosis of Paget's disease. [Jun 08]**
64. **Discuss the techniques, imaging features & limitations of sonographic evaluation of the rotator cuff [Jun 08]**
65. **Describe different types and imaging features of fractures. What are the complications of fracture? [2010]**
66. **Define Stress fracture. Enumerate various sites and predisposing factors of stress fracture. Describe various imaging features of stress fractures. [Dec 10]**
67. **Describe in brief various imaging features of Osteoid osteoma. Discuss its differential diagnosis. [Dec 2010]**
68. **Describe the life cycle of hydatid disease causative organisms. Enumerate sites of affection in human beings. Describe imaging features of Musculoskeletal hydatidosis. [Dec 2010]**

69. Enumerate causes of Hypertrophic osteoarthropathy. Discuss its DD & describe its imaging findings on plain radiograph. [Jun 2011]
70. Discuss the radiographic and sonographic features of developmental dysplasia of Hip. [Jun 2011]
71. Describe the MR anatomy of the knee joint. Briefly state the MR sequences you would employ to delineate a suspected medial meniscus tear. [Jun 11]
72. Enumerate different varieties of Osteosarcoma. Discuss their imaging features. [3+7 Dec 11]
73. Describe etiopathogenesis of Osteomyelitis. Discuss role of imaging in acute osteomyelitis. [4+6 Dec 11]
74. Discuss the clinical associations of Hypertrophic Osteoarthropathy. Briefly describe its radiological findings, Differential diagnosis and role of Nuclear medicine. [3+4+2+1 Dec 11]
75. Classify scoliosis. Discuss imaging features of plain radiographic, CT and MRI in neurofibromatosis of spine. Discuss Cobb's angle and draw a diagram illustrating its measurement. [2+5+2+1 Jun 12]
76. Mention causes of periosteal new bone formation. Briefly discuss characteristic radiological features of osteomyelitis affecting infants, children and adults. [2+8 Jun 12]
77. Mention differential diagnosis of 15 year boy presenting with localized pain and swelling of 2 months duration in right lower thigh. Discuss conventional radiographic, CT and MRI features of the commonest primary malignant bone tumor in this age. [1+3+3+3 Jun 12]
78. Classify cysts of jaw. Describe briefly imaging features of each type of cyst. Draw suitable diagrams to describe various types. [2+6+2 Jun 12]
79. Discuss differential diagnosis and imaging features of painless expansile lesion involving single rib in an adult. [3+7 Jun 12]

80. List the causes of posterior scalloping of vertebrae. Describe skeletal changes seen in von Recklinghausen's disease. [2+8 Dec 12]
81. Enumerate various causes of hemolytic anemia. Describe the imaging findings in a case of Thalassemia major. Briefly discuss its DDs from sickle cell anaemia. [2+5+3 Dec 12]
82. Briefly discuss the pathophysiology of osteomalacia. Describe the radiological findings in renal osteodystrophy. Enumerate the findings that help in differentiating from primary hyperparathyroidism. [3+4+3 Dec 12]
83. What is Osteoporosis? Enumerate causes of osteoporosis. Discuss any 3 imaging modalities currently in vogue for assessment of bone mineral density. [2+2+6 Jun 13]
84. What are the key clinical features, common sites and radiological findings in Ewing's sarcoma? Discuss its differential diagnosis in brief. [2+2+4+2 Jun 13]
85. The Child Welfare Board has referred an accused to you for estimation of age. Being a radiologist, how would you carry out this assignment?. Discuss in brief the variables that can affect the estimated age. [6+4 Jun 13].
86. Enumerate causes of painful limp in a child unable to bear weight. Briefly discuss the role of plain X-ray, arthrography, US, CT, MRI and scintigraphy in arriving at diagnosis. [2+2+1+1+1+2+1 Jun 13]
87. Discuss the role of plain X-ray, CT and MRI in cases of lower cervical spinal trauma. [3+4+3 Jun 13]
88. Describe the MR anatomy of the shoulder joint. Briefly state the MR sequences you would employ to delineate various lesions of the shoulder joint. [4+6 Jun 13]
89. a. Ossification of elbow joint and its clinical significance. b. Fusion imaging. [5+5 Jun 13]
90. Enumerate any 5 morphological patterns of periosteal reaction and state their clinical significance. [2+2+2+2+2 Dec 13]
91. Discuss the pathophysiology of osteomalacia. Describe imaging features in primary hyperparathyroidism. [5+5 Dec 13]

92. Enumerate the hematopoietic disorders which causes marrow changes. Discuss the MRI findings of any two of these marrow disorders. [2+4+4 June 14]
93. Discuss the imaging features of avascular necrosis of the hip and its DD. [7+3 June 14]
94. What are the causes and imaging features of hypertrophic osteoarthropathy [3+7 June 14]
95. A 10-yr-old child has presented with swelling of the mandible. Enumerate the causes and discuss the imaging findings of any two. [2+2+6 June 14]
96. Enumerate the causes of hypertrophic osteoarthropathy. Briefly describe its radiological findings, DD and role of Nuclear medicine. [2+4+2+2]
97. Enumerate various causes of hemolytic anemia. Describe the imaging findings in a case of Thalassemia major. Briefly discuss its DDs from sickle cell anaemia. [2+5+3 Dec 14] (*this question was repeated from Dec 12*)
98. List the causes of posterior scalloping of vertebrae. Describe skeletal changes seen in von Recklinghausen's disease. [2+8 Dec 14] (*repeat from Dec 12*)
99. 99. Enumerate causes of painful limp in a child unable to bear weight. Briefly discuss the role of plain X-ray, arthrography, US, CT, MRI and scintigraphy in arriving at diagnosis. [2+2+1+1+1+2+1 Dec 14](*exact repeat from June 13*)
100. Classify cysts of jaw. Describe briefly imaging features of each type of cyst. Draw suitable diagrams to describe various types. [2+6+2 Dec 14](*exact repeat from June 12*)
101. a. Ossification of elbow joint and its clinical significance. b. Fusion imaging. [5+5 Dec 14] (*exact repeat from Jun 13*)
102. a) Enumerate different varieties of osteosarcoma. b) Imaging features of various surface osteosarcomas. [5+5 June 15].
103. a) Pathophysiology of different types of hyperparathyroidism. b) Imaging features of primary HPT [5+5 June 15].
104. a) MRI anatomy of knee joint. b) Role of MRI in evaluation of meniscal injuries.

105. Various osseous changes in NF. [June 15]
106. Causes of paravertebral shadow in lumbar region and their differential diagnosis. [June 15]
107. a) Enumerate causes of periosteal new bone formation. b) Imaging features of acute osteomyelitis and infantile cortical hyperostosis. [3+(4+3) Dec 15]
108. Various radiological findings in Battered Baby Syndrome and their mimickers. [6+4 Dec 15]
109. Radiological findings in: a) Dermatomyositis. b) Lipoma arborescens. [5+5 Dec 15]
110. a) Role of skeletal radiography in determination of bone age in a male likely to be between 12-18 years of age. b) Factors affecting bone growth and remodeling. [6+4 Dec 15]
111. a) Radiological features of osteopenia on a plain radiography. b) Various imaging modalities used in the assessment of bone mineral density. [3+7 Apr 16]
112. Various radiological findings in multiple myeloma and its differential diagnosis. [6+4 Apr 16]
113. Radiological findings in primary hyperparathyroidism. [10 Apr 16].
114. a) Enumerate the various mucopolysaccharidosis. b) Radiographic findings in Morquio syndrome. [2+8 Apr 16]
115. Radiological evaluation of skeletal maturity for bone age estimation. [10 Apr 16]
116. What are the varieties of osteosarcoma? Enumerate the imaging features. [2+8 Dec 17]
117. What are the clinical features in a patient with rheumatoid arthritis? Describe the various imaging characteristic in rheumatoid arthritis? [4+6 Dec 17]
118. Define stress fracture. Enumerate the common sites. Discuss the role of imaging in early diagnosis. [2+2+6 Dec 17]
119. Enumerate the cystic lesions of jaw. Outline the radiological algorithmic approach towards the diagnosis of such a lesion. [4+6 Jun 18]

120. Enumerate the various labral injuries of shoulder.
Describe the magnetic resonance imaging findings in any two such injuries. [2+4+4 Jun 18]
121. Enumerate the causes of periosteal reaction in childhood.
Discuss the radiological approach towards its diagnosis. [3+7 Jun 18]
122. State the age of appearance and fusion of various epiphyses around the elbow joint on plain radiographs in a male subject. Discuss how radiography can help in gender determination in an adult subject. [5+5 Jun 18]
123. a) Explain the role of plain radiography in various diseases of the joints. b) What is the role of ultrasound of joints in the evaluation of a case of rheumatoid arthritis? [5+5 Dec 18]
124. Enumerate the causes of osteoporosis. Discuss the role of plain radiography in the evaluation of skeletal manifestations of hyperparathyroidism. [4+6 Dec 18]
125. Role of imaging in short stature. [10 Jun 19]
126. What are the various forms of LCH? Describe the radiological features of the imaging spectrum. [3+7 Jun 19]
127. Which are the common sites for occurrence of Brown tumours? How will you evaluate the patient to detect the underlying abnormality? [4+6 Jun 19]
128. a) Enumerate various causes of hemolytic anemia. b) Describe the imaging findings in a case of thalassemia major. How will you differentiate it from sickle cell disease on imaging? [2+5+3 Dec 19]
129. a) Describe in brief anatomy of the shoulder joint. b) Briefly state various MR sequences and planes used to evaluate shoulder joint on MRI. [4+6 Dec 19, *Repeat from June 13*]
130. A 15-year boy presents with localized swelling and pain of 2 months duration involving right lower thigh. a) What will be its differential diagnosis? b) Discuss Conventional Radiographic, CT and MRI features of the commonest primary malignant bone tumour in this age. [4+6 Dec 19, *Repeat from June 12*]

NEURORADIOLOGY

1. CT in Neurotuberculosis.
2. Discuss the role of computed tomography in cerebrovascular accidents.
3. Ring lesions of brain
4. Neurosonography.
5. Discuss the role of computed tomography in infective lesions of brain.
6. Radiology and Imaging in Acoustic Neuroma.
7. Radiological diagnosis of extradural spinal masses.
8. Enumerate the various neurocutaneous syndrome & describe imaging in any 2 of these.
9. C.T. versus M.R.I. in brain tumours.
10. Imaging in congenital lesions of the spine and spinal cord.
11. Radiology and Imaging of Meningiomas.
12. CT and MRI in Intervertebral disc Prolapse.
13. Discuss the anatomy and anomalies of the Cranio-Vertebral region. How will you assess it Radiologically?
14. White matter disorders OR CT and MRI in white matter diseases OR MRI in CNS white matter disease of Brain.
15. 4th Ventricular Ependymoma.
16. Migrational anomaly.
17. Role of DWI in brain .
18. Sub Arachnoid space anatomy and SAH and intervention.
19. Intervention in SAH.
20. Posterior fossa neoplasms of childhood.
21. Lateral ventricular masses.
22. Radiology of brain tumors.
23. Role of Imaging in Leukodystrophies.
24. Carotico-Cavernous Fistula.
25. Imaging and Intervention in Intracranial and spinal AVM.
26. MRI in Alzheimer's disease.
27. Radiological features in diffuse axonal injury.
28. Central pontine myelinolysis.
29. Vein of Galen malformation.

30. Glutaric Aciduria Type I.
31. MRI in intramedullary neoplasms of spinal cord.
32. Clinical applications of Diffusion Tensor imaging.
33. Embolization in management of acute hemorrhage.
34. Neuro-imaging in AIDS.
35. Radiological features in Neurological Complications of AIDS.
36. Functional Imaging of the Brain.
37. Enumerate CP angle tumors and discuss their differentiating features on CT and MRI.
38. Describe MR anatomy of pituitary gland. Discuss in detail MR techniques and features to diagnose pituitary adenomas.
39. Define Spinal Dysraphism. Describe briefly the MR sequences you will use for diagnosis of spinal dysraphism. [2010]
40. Define Infective Discitis. Describe CT, MRI and Isotope imaging features of discitis. [2010]
41. Enumerate various causes of Supra-sellar masses. Describe imaging features in Craniopharyngioma.
42. Describe MR Imaging features in intra-cerebral Hematoma and techniques of Volume calculation in such a case.
43. Classify neural tube closure defects. Describe various Chiari malformation and their imaging features.
44. Classify brain tumors of children. Describe the imaging features of Primitive Neuroectodermal tumors. [Dec 2010]
45. Describe MR anatomy of Pituitary fossa with a diagram. Describe the radiological diagnosis of Pituitary adenoma. [Dec 2010]
46. Describe the clinical features and MR imaging features of Multiple Sclerosis.
47. Describe the grading, imaging features and differential diagnosis of Glioblastoma multiforme.
48. Describe the CT and MRI features of Neurocysticercosis of the brain. How would you differentiate from other granulomatous lesions?
49. Discuss the role of CT in evaluation of patients with acute head injury, staging the types of injury, possible complications and their long term sequelae.

50. Enumerate the clinical uses of MR spectroscopy in disorders and diseases of the brain . Briefly discuss its role in evaluation of brain tumors.
51. Enumerate causes of demyelinating diseases of the spinal cord. Discuss their imaging features and DDs.
52. Enumerate causes of normal intracranial calcifications. Discuss imaging features of pathological intracranial calcifications secondary to infections and infestations.
53. Enumerate causes of spinal canal stenosis. Mention normal CT measurement of spinal canal at various levels. Describe plain radiographic, CT and MRI features of spinal canal stenosis.
54. A 38 year old man, who has been throwing epileptic seizures, is found to have ring lesions on MRI of the brain. Discuss the differential diagnosis. Describe the specific MRI features of any 4 clinical entities which may present with these clinikoradiological findings.
55. State the distinguishing features of intramedullary, extramedullary, intradural and extradural spinal lesions on MRI. Discuss briefly the DD's of intramedullary spinal lesions.
56. How would you differentiate between an extra-axial and intra-axial mass lesion on cranial MRI? Describe the radiological findings in the most common extra-axial lesion found in middle aged patients.
57. Discuss the types and classification of gliomas. Describe the imaging features of various types of gliomas. Discuss the role of perfusion imaging in gliomas.
58. What is the basic difference b/w NF type I and II? Discuss the imaging findings in NF-II. Briefly describe extra skeletal manifestations and associations of Neurofibromatosis.
59. Describe with a diagram the cerebral venous system. Briefly discuss the causes and imaging features of cortical venous thrombosis
60. A 40-yr-old female has presented with loss of vision and instability in gait. Discuss the DD and MRI findings in the most probable cause. What is the role of diffusion tensor imaging in this patient.

61. a) Enumerate various neurocutaneous syndromes. b) Etiopathogenesis imaging features and associations of Sturge-Weber syndrome.
62. 91. Cystic lesions of posterior fossa and their differential diagnosis.
63. 92. a) Enumerate the causes of epilepsy. b) Distinguishing features of cortical lesions associated with epilepsy
64. What are the causes of hypoxic Ischemic encephalopathy in children and adults? Enumerate imaging features.
65. Discuss the role and imaging features of MRI in multiple sclerosis.
66. Enumerate the various types of primary traumatic lesions in head injury. State the indications for CT and that for MRI in head trauma.
67. Enumerate the possible causes of a jugular foramen mass. Describe the magnetic resonance imaging features in any two such lesions.
68. Enumerate the possible causes of a sudden unilateral loss of vision. Discuss the role of a radiologist in the diagnosis of any two such conditions which may present in this manner.
69. Enumerate the various causes of bilateral T2 hyper intense lesions in basal ganglia. Describe the neuroimaging findings in Wilson's disease.
70. Draw a labelled diagram of a coronal section through the "cavernous sinus". b) Discuss the differential diagnosis of cavernous sinus lesions.
71. a) Describe in brief revised WHO classification of brain tumours. b) Enumerate primitive Neuroectodermal tumours of the brain.

NUCLEAR MEDICINE

1. Radio nuclide imaging of the C.N.S.
2. Radio isotope scanning in thyroid disease.
3. Isotope imaging of the Parathyroids.
4. Radio nuclide imaging of urinary tract.
5. Isotopes in Myocardial ischaemia OR Scintigraphy in ischaemic Heart disease OR Role of nuclear medicine in ischaemic heart disease

6. Clinical application of radionuclide Renography.
7. Application of DMSA Scintigraphy.
8. Scintigraphy evaluation of Gastro-intestinal bleeding.
9. Fusion imaging.
10. Describe Radiopharmaceuticals used in PET CT with their clinical applications.
11. Enumerate various radio-isotopes used in Hepato-Biliary system. Describe the imaging features and techniques in Biliary atresia.
12. Describe the role of scintigraphy in cardiac imaging with emphasis on myocardial perfusion and viability.
13. Enumerate the indications of scintigraphic evaluation in GI bleed. Briefly discuss technique, radioisotopes used and interpretation of results.
14. Enumerate indications and radio-isotopes used for radionuclide scanning of lungs. Briefly describe 3 techniques of isotope imaging of lung with their clinical implications.
15. a. Radio isotope scanning of the skeletal system. b. Clinical applications of 3D and 4D ultrasound.
16. Discuss the role of scintigraphy in cardiac imaging with special emphasis on myocardial perfusion and viability. [10 Jun 13]
17. What is the principle of PET scanning? Briefly discuss the role of FDG-PET scanning and importance and clinical utility of two non-FDG molecules of PET scanning.
18. a) Renal isotope scanning b) Tomosynthesis in mammography.
19. Principle and role of PET scanning in benign conditions affecting the lymph nodes.
20. Discuss the indications, pharmacokinetics, and method of iodine-131 metaiodobenzylguanidine scintigraphy.

OBSTETRICS AND GYNECOLOGY

1. Imaging of IntraUterine Foetal Death.
2. Discuss the role of imaging in uterine lesions.
3. Endometriosis.
4. Alimentary tract lesions diagnosable in-utero
5. Role of Sonography in I.U.G.R. Mention the significance of the aortic isthmus index.
6. PCOD.
7. Imaging in Infertility.
8. Biophysical score.
9. Uterine interventions
10. PCPPNDT
11. MRI in gynecologic imaging.
12. Cystic lesions of ovaries.
13. Radiological evaluation of delayed milestones.
14. Antenatal detection of Vein of Galen malformation.
15. Antenatal MRI.
16. CT – Pelvimetry.
17. Enumerate markers of chromosome abnormality on antenatal ultrasound. Briefly discuss their sonographic features.
18. Uterine artery embolisation.
19. Describe the sonographic findings favouring the diagnosis of ectopic pregnancy and its DD.
20. Enumerate the vascular and structural abnormalities of the Umbilical cord. Describe the velocity waveform changes seen in the umbilical artery Doppler.
21. What is the role of imaging in assisted reproduction?
22. Define fetal hydrops. Enumerate its causes. Describe sonographic and color doppler findings noted in this condition.
23. Discuss the sonographic techniques and criteria used in evaluation of uterine cervical incompetence.
24. Outline the diagnostic imaging approach in a pt. with Ovarian malignancy. Describe imaging features, staging & impact of cross sectional imaging in ovarian cancer. Describe the role of imaging in recurrence of ovarian malignancy after surgery.

25. Define abnormal endometrial thickening. Enumerate its causes and discuss their imaging features.
26. List various causes of bleeding in the first trimester. Discuss their sonographic features.
27. Describe measurement technique & normal values of nuchal translucency. Briefly discuss its role in Trisomy 21 and other chromosomal anomalies.
28. List various indications of MRI examinations in obstetrics. Outline various sequences used along with their rationale. Briefly describe MRI findings in two conditions presenting obstetric emergencies.
29. List the causes of post menopausal bleeding. Briefly describe the role of various imaging modalities highlighting their advantages and pitfalls. Describe the MRI findings in a case carcinoma cervix.
30. A 19 year old girl is referred with complaint of primary amenorrhea from the Department of Gynecology. As a radiologist, how would you evaluate her? Enlist the radiological investigations that might be beneficial to her, enumerating the precise entities you might identify with each. Describe the key radiological findings in any one clinical condition which may present as primary amenorrhea.
31. A 26 year old patient, who is 12 weeks post-partum, is referred to you for radiological appraisal with a history of bleeding per vaginum and raised beta-HCG levels. What is the likely diagnosis?. How would you evaluate this patient?. Discuss the possibilities with their key radiological findings.
32. How would you decide on the amnionicity and chorionicity in twin pregnancies? Enumerate the various complications that may occur in a twin pregnancy. Describe the various radiological findings in twin-twin transfusion syndrome.
33. Enumerate and classify various congenital anomalies of the uterus. Discuss the role of US and MRI in their diagnosis, highlighting their advantages and limitations.
34. Define habitual abortion. Enumerate various causes of habitual abortion. Discuss the role of imaging in diagnosis and follow up of these cases.

35. Ultrasound in a 35 yr old female shows a right adnexal cystic mass. a) What are the likely causes? b) Algorithmic approach and imaging features in these causes.
36. a) Normal anatomy of placenta. b) What are the different types of abnormal placental implantation. c) Role of imaging in placenta accreta. a) Documentary requirements under PC-PNDT Act. b) Positioning and technique for Water's view.
37. Imaging findings of ovarian torsion. Role of MRI in
38. Current indications and limitations of Foetal MRI.
39. Enumerate the causes of a unilateral adnexal mass. Discuss the role of a radiologist in diagnostic evaluation of such a patient.
40. a) Intrauterine image-guided interventions which can be carried out in pregnancy. b) Role of Doppler ultrasound in the evaluation of foetal middle cerebral artery.
41. Describe the role of ultrasound, computed tomography and magnetic resonance imaging in the diagnosis and staging of endometrial carcinoma.
42. Imaging findings in molar pregnancy.
43. Role of Doppler in pregnancy.

PANCREAS

1. Endocrine tumours of the pancreas.
2. Role of imaging in acute pancreatitis.
3. Pancreatic Endosonography.
4. Image guided interventions in pancreatic disease.
5. Radiological features in cystic tumors of pancreas.
6. What is Pancreatic divisum?. Briefly discuss its embryologic basis and clinical significance. What are ERCP, MRCP and MDCT findings.
7. Enumerate various pancreatic masses of childhood. Discuss their imaging features and DDs of pancreatoblastoma.
8. Briefly describe embryological development of pancreas. Describe various anomalies and variations in its development with the help of

suitable diagrams. Discuss imaging features (on barium meal and CT scan) of annular pancreas.

9. Enumerate various indications of upper gastrointestinal endoscopic sonography. Briefly discuss its role in evaluation of pancreatic pathologies outlining the advantages and disadvantages.
10. A 40 year old female with pain abdomen is found to have a cyst in the body of pancreas on USG. Enumerate various possible causes. Discuss the imaging algorithm you would follow for arriving at diagnosis in this case.
11. Anomalous pancreatobiliary ductal junction and its complications
12. Intraductal papillary tumors of pancreas.
13. b) Biliary atresia
14. a) Enumerate various neuroendocrine tumors of pancreas. b) Characteristic features of these on various imaging modalities including the role of radio-nuclide imaging.
15. a) Imaging and interventions in vascular complications of pancreatitis. b) Imaging features of multicystic dysplastic kidney.

PEDIATRICS

1. Enumerate the causes of presacral masses in children. Discuss the various imaging features of each.
2. Role of plain X-ray of wrist in determining the age of child in medicolegal cases.

PHYSICS

1. Technical parameters of an x-ray equipment for fluoroscopic procedures.
2. Basic construction of an x-ray tube and recent advances.
3. Principle of doppler ultrasound and its application in neck ultrasound.
4. Factors affecting quality of a radiograph.
5. Name the various interactions of X-ray photons with matter. Describe any two.
6. Focal spot in a diagnostic x-ray tube.
7. Ultrasound image artifacts.

8. Image Intensifier.
9. Discuss the Biological effects of Radiations and the measures taken against its protection for Radiation workers and patients in Radio-diagnosis dept.
10. Define principles of radiation protection. Describe various parameters which can reduce patient radiation dose in radiography and fluoroscopy.
11. AERB guidelines for Radiation safety.
12. Ionizing radiation in bone
13. Portable radiography
14. MDCT technology.
15. Construction of a conventional X-ray film, its types and functions of each layer.
16. Discuss about different parameters which influence film contrast.
17. Properties of X-rays.
18. Medical X ray films processing chemicals.
19. High generator transformer.
20. Radiation monitoring devices.
21. Define scatter radiation. Discuss briefly the parameters which influence scatter radiation and methods to reduce scatter radiation.
22. Rare earth screens.
23. Cine fluoroscopy
24. Grids
25. Cardiac CT.
26. Darkroom illumination.
27. Modern rotatory x-ray tube .
28. PACS – picture archival and communication system.
29. TLD – Thermo Luminescence Dosimeter .
30. MR coils.
31. Design and setup of a radiology department OR Setting up a radiology department in a 200 bedded hospital
32. Film artifacts.
33. Electrical circuits of x-ray machine .
34. Safety hazards in MRI.
35. Steps to improve the quality of a chest X-ray .

36. Radiological management of Bomb-Blast injury.
37. Maximum permissible radiation dose.
38. Computed radiography cassette.
39. PACS in radiology.
40. Genetic Screening
41. Define basic units of radiation exposure. List recommended dose limits for radiation workers and the general public.
42. Discuss about the mammography X-ray unit.
43. Legal responsibilities and duties of radiologist in clinical practice.
44. Define quality assurance. Discuss the organization of a quality assurance program pertaining to radiology equipment.
45. Principles and clinical applications of dual energy CT.
46. Describe the basis of BOLD imaging. Write its utility and limitations.
47. Write in brief the principle and types of Digital radiography. Outline its advantages and disadvantages.
48. Describe AERB guidelines on X-ray room installation.
49. Brief outline the evolution of present day CT scanners citing the key specific changes through different generations.
50. While conducting a conventional diagnostic radiographic procedure under fluoroscopic guidance , what steps would you take to reduce radiation dose to pt. what measures would you take to safeguard yourself.
51. Discuss briefly the principle of MR spectroscopy. Enumerate its clinical significance in any three clinical settings , outlining explicitly how it would be useful.
52. Discuss various dose reduction techniques in MDCT. Mention the average radiation doses received for common examination using MDCT.
53. Discuss various statutory requirements to be followed for installation of following radiological equipments:
 - A. 1000 mA x-ray machine
 - B. CT scan
 - C. DSA Lab
54. Conventional lead apron and zero lead apron
55. Write short notes on the following:
56. Green sensitive film and Dual energy substractions.

57. a) Personal Dosimeters b) Tissue Harmonic imaging.
58. a) MR contrast for liver imaging b) Contrast induced nephropathy
c) What will you do to prevent CIN?
59. Advances in CT technology to decrease the radiation dose in children. What is CT dose index (CTDI).
60. Write in brief the principle and types of Digital radiography. Outline its advantages and disadvantages.
61. Clinical applications and techniques of fat suppression in MRI.
62. Principles of perfusion CT and quantification of tumor perfusion parameters.
63. Enumerate the types of digital radiography. Describe each one briefly.
64. b) Advantages and limitations of digital radiography compared with conventional film screen radiography.
65. Describe the physical principle behind various diffusion weighted imaging techniques. Discuss their clinical usefulness.
66. Enumerate the constituents of ultrasound jelly. State the significance of each constituent.
67. Enumerate the various pulse sequences used in MRI. State the clinical utility of each pulse sequence citing examples.
68. What is MRI fringe field? How can the MRI safety risks be minimized in your department?

RADIOGRAPHIC POSITIONING

1. Describe the positioning for various skull x-ray views.
2. Conventional skull radiography.
3. Radiography of the Jugular Foramen.
4. Base of Skull.
5. a) Positioning and technique for apicogram. b) Magnification radiography.

SKULL AND ORBIT

1. Investigation in a case of exophthalmos.
2. Imaging of posterior fossa.
3. Orbital tumours.
4. USG in retinal & choroidal detachment.
5. Orbital pathologies.
6. Ocular blood flow in normal and Glaucomatous eye on color Doppler imaging.
7. Discuss MR features in orbital pseudotumor.
8. Enumerate causes of unilateral proptosis. Describe imaging findings of optic glioma and carotid cavernous fistula.
9. Describe in brief anatomy of sella turcica. Enumerate various sellar and parasellar masses. Discuss imaging features of craniopharyngioma.
10. Enumerate various indications of orbital ultrasound. Discuss the role of ultrasound and color Doppler in a case of white reflex in a child.
11. Enumerate the cause of solitary lytic lesion in the skull. Describe the distinguishing radiological features of any three.
12. Enumerate the different extraconal lesions of the orbit. Briefly mention their imaging findings.
13. Radiological assessment of CSF rhinorrhea.
14. Describe the anatomy of cranio-vertebral junction with the help of diagram(s). What are the various radiological measurement parameters to diagnose Basilar Invagination?

TECHNIQUES, NEWER MODALITIES AND RECENT ADVANCES

1. Spiral CT and its major applications
2. Ultrasound transducers and their applications.
3. Developments in ultrasound transducer technology.
4. Xeroradiography.
5. High resolution CT and its major applications.
6. MR Spectroscopy.
7. Automatic processing and Automatic Film Processor (AFP).
8. 3D CT angiography.

9. Flat panel digital radiography.
10. What is digital radiography? Discuss its advantages and disadvantages.
11. MRCP vs ERCP
12. Small bowel enema.
13. Tissue Harmonic imaging.
14. MRI Urography.
15. MR Venography.
16. MR angiography , MRA in lower limb arteries
17. Discuss the role of C.T. angiography, its principles indications, advantages and limitations, its application in abdomen and its present status.
18. Methods of contrast administration for CT angiography.
19. CT angiography vs MR angiography.
20. Virtual endoscopy.
21. Virtual Colonoscopy.
22. Virtual bronchoscopy .
23. CT Coronary angiography.
24. Intra-operative USG.
25. Trans-rectal and Trans-Perineal USG in elderly patients.
26. Sonohysterography.
27. Discuss principle, various techniques of elastography and their clinical applications.
28. Peripheral venous doppler.
29. DSA.
30. Intravascular Ultrasound.
31. Full field Digital Mammography.
32. Radiofrequency Ablation.
33. Percutaneous vertebroplasty.
34. Outline of techniques in functional MRI.
35. Diffusion weighted MRI.
36. Dry view laser camera.
37. Discuss the procedure for Barium Enema.
38. Technique of Double Contrast Barium Enema.
39. MR enteroclysis – techniques, indications and applications.

40. MRI in Cardiac Imaging OR MR sequences in Cardiac Imaging.
41. Vertebroplasty in non-infective vertebral collapse.
42. Tomosynthesis and its clinical applications.
43. Volume ultrasound.
44. Discuss indications, technique and complication of bronchial artery embolisation.
45. PACS.
46. Describe the principle and types of bone densitometry. Outline the advantages, disadvantages and limitations of each type.
47. Enumerate the various gradient echo sequences. Describe in brief the principle and their clinical applications.
48. Techniques and applications of CT colonography.
49. What are the advantages of 3T MRI over 1.5T MRI ? Comment on its limitations.
50. What is the principle of diffusion weighted imaging and its role in evaluation of breast masses?
51. Enumerate the indications of foetal MRI. Comment on its limitations.
52. Describe briefly indications, technique, complications and post procedure follow up of Transjugular Intrahepatic Portosystemic Shunt.
53. What is molecular imaging and describe its role in the musculoskeletal system.
54. Describe the technique of MR Arthrography. Enumerate its indications, advantages and limitations.
55. What do you understand by perfusion imaging?. Describe briefly CT and MR perfusion imaging techniques.
56. Radiofrequency ablation in hepatic and biliary lesions.
57. Define High Intensity Focused Ultrasound. Describe its clinical applications.
58. Describe principle of Dual energy CT, different techniques of dual energy acquisition and various applications.
59. Describe the physical principles of PET-CT. Discuss the role of computed tomography and PET-CT in diagnosis post-treatment evaluation of lymphoma.

60. Describe the physical principles of CR and DR. Briefly discuss their advantages and disadvantages.
61. 81. Write short notes on: A. Focussed abdominal sonography for trauma. B. Pressure injector
62. Define strain and shear wave elastography. Discuss its role in breast, prostatic and musculoskeletal lesion. Compare its sensitivity and specificity with MR elastography.
63. a. MR artefacts b. CT artefacts.
64. Discuss the following: a. BOLD Imaging b. Genetic Screening c. PACS in Radiology.
65. Write short notes on: 1) MR tractography 2) Dual energy scanning in musculo-skeletal system.
66. Write short notes on: 1) CO2 angiography 2) Transcranial sonography in stroke.
67. Write short notes on: a) Flat panel detector b) HIFU-Clinical indications & utility.
68. Write short notes on: a) Mobile CT scanner b) Renal denervation for renovascular hypertension.
69. a) MR-PET b) Dose reduction techniques in MDCT.
70. a) Imaging of hemobilia and interventions b) Principles & applications of RF ablation.
71. a) CT perfusion in acute stroke b) Principles of functional MRI.
72. a) Fusion imaging. b) ELORA.
73. Advances in ultrasound transducer technology.
74. a) Zero lead aprons. b) Spatial compound imaging
75. a) Advances in lead apron technology. b) Technique of MR perfusion and its clinical applications in brain lesions.
76. a) MR spectroscopy in breast. b) MR spectroscopy in the prostate.
77. a) PET-MR b) MR guided interventions.
78. Role of perfusion and diffusion MRI in post chemotherapy evaluation.
79. What is enteroclysis? Discuss its technique with merits and demerits with Barium meal follow through.
80. Principle, equipment, uses and adverse effects of RFA.

81. Discuss the pre-procedure work-up, procedure, and the post-procedure care in a patient scheduled to undergo a trucut/core biopsy for a hepatic mass lesions.
82. Describe the indications, contraindications and technique of CT urography.
83. Define 4DCT technique and enumerate the situations in which it is used and discuss in detail the technique of 4DCT of parathyroid.
84. Discuss the factors which tend to affect the image quality on a CT scan. Describe the diverse radiation dose reduction strategies which can be employed while carrying out a thoracic CT scan.
85. Enumerate the various post-processing techniques which can be used during a thoracic CT examination. State the clinical utility of each of these post-processing techniques citing examples.
86. Describe the basic principles of PET-CT. Discuss the advantages and disadvantages of PET-MRI over PET-CT.
87. Discuss the various techniques of cartilage imaging. Describe the common sites and imaging features of osteochondral lesions.
88. Role of radiology in diagnosing and treating renovascular hypertension.
89. Describe the AERB guidelines to set up a Radiology centre for a single static X-ray machine. Supplement it with a diagram.
90. a) What do you mean by AERB, eLORA and ALARA? b) How will you minimize Radiation Hazards?
91. a) Artificial Intelligence in Health care. b) Discuss its advantages and disadvantages.

THYROID

1. Role of USG in thyroid diseases.